APPENDIX

1

Reported Sizes and Sequences of STR Alleles

Explanation of information included in the following tables:

This appendix material describes the reported alleles for the 23 short tandem repeat (STR) loci most commonly used in the United States and around the world. Note that the number of alleles present for a particular locus is an indication of the polymorphic nature of that marker and its value for use in human identity testing. The STR locus FGA is more variable than TPOX because it possesses more alleles and thus there is a greater chance that two individuals selected at random would have different genotypes at FGA than at TPOX. Of course, the frequency at which alleles occur in a population plays an important role in the effectiveness of STR loci to distinguish between individuals. Allele frequency information for these 23 STR loci will be included in the forthcoming volume *Advanced Topics in Forensic DNA Typing: Interpretation*.

As described in Chapter 5, STR alleles are named based on the number of full repeat units that they contain while partial repeats (i.e., microvariants) are designated by the number of full repeats, a decimal, and the number of nucleotides present in the partial repeat in accordance with International Society of Forensic Genetics (ISFG) recommendations (1994). Some of the alternative allele nomenclatures that were present in previous editions of *Forensic DNA Typing* have been removed. Alleles that are the same overall length but possess a different known internal sequence structure are sub-labeled below as "(a)," "(b)," etc. (e.g., FGA 27(a) and FGA 27(b)). Adjacent rows containing same-size, different-sequence alleles are shaded in gray.

Known flanking sequence differences that alter the DNA sequence length or electrophoretic migration of a PCR product are designated according to the 2006 ISFG recommendations with additional information after the number of complete STR repeat units (Gusmão et al. 2006, Butler et al. 2008). For example, a TPOX allele with 8 repeats and a single deletion of an adenine ("A" nucleotide) 48 bp downstream of the repeat is designated 7.3 due to its migration (Allor et al. 2005) or 8(D48Adel) to describe its flanking sequence information, where 8 stands for the number of complete repeats, D48 indicates the direction

and position of the mutation relative to the STR repeat block (i.e., the mutation is located 48 bases downstream of the repeat), and "Adel" indicates that an A nucleotide has been deleted.

The polymerase chain reaction (PCR) product size for each of the possible alleles is listed based on the commercial STR kit used to amplify the particular locus. Commercially available STR kits can produce different DNA fragment sizes because their primers hybridize to different positions in the flanking regions of the STR sequence. The expected DNA fragment sizes (based on their actual sequence) for all reported alleles are calculated for the designated Promega PowerPlex kit or the Applied Biosystems AmpFlSTR kit. These PCR product sizes are listed without any nontemplate addition, i.e., they are in the "−A" rather than the "+A" form. *STR allele sizes measured in a laboratory may also vary from the actual sequence-based size listed here due to the internal sizing standard used and the particular electrophoretic conditions.* In addition, some of the Applied Biosystems loci have mobility modifiers to alter the relative size of the PCR products during electrophoretic measurement.

The common repeat sequence motif for each STR locus is listed according to the 1997 ISFG recommendations (Bär et al. 1997). In most cases, the sequence changes in the repeat region are the only variation occurring and the flanking sequences remain constant. However, variation in the flanking sequence is also a possibility as has been shown with base composition mass spectrometry studies (Oberacher et al. 2008). Finally, the reference is listed where each new allele (and its sequence if published) has been described.

New (rare) alleles will be discovered as more samples are analyzed using these STR loci. As this listing becomes outdated, readers are encouraged to consult the STRBase variant allele listing (http://www.cstl.nist.gov/biotech/strbase/var_tab.htm) and to contribute newly discovered alleles so that they may be categorized for fellow workers in this field. The complete sequence for one of the alleles listed for each locus may be found by using the GenBank accession number listed for that locus or by checking the reference sequence in STRBase (http://www.cstl.nist.gov/biotech/strbase/seq_info.htm).

CSF1PO

GenBank Accession: X14720 (allele 12). PCR product sizes of observed alleles

Allele (Repeat #)	Promega PowerPlex 16	ABI Identifiler	Repeat Structure $[AGAT]_n$	Reference
5	317 bp	301 bp	$[AGAT]_5$	Kline et al. (2010)
6	321 bp	305 bp	$[AGAT]_6$	Lazaruk et al. (1998)
6.3	324 bp	308 bp	Not published	STRBase
7	325 bp	309 bp	$[AGAT]_7$	Huang et al. (1995)
7.3	328 bp	312 bp	Not published	STRBase
8	329 bp	313 bp	$[AGAT]_8$	Puers et al. (1993)
8.3	332 bp	316 bp	Not published	STRBase
9	333 bp	317 bp	$[AGAT]_9$	Puers et al. (1993)
9.1	334 bp	318 bp	Not published	STRBase
9.3	336 bp	320 bp	Not published	STRBase
10	337 bp	321 bp	$[AGAT]_{10}$	Puers et al. (1993)
10.1	338 bp	322 bp	Not published	Huel et al. (2007)
10.2	339 bp	323 bp	Not published	Allor et al. (2005)
10.3	340 bp	324 bp	Not published	Lazaruk et al. (1998)
11	341 bp	325 bp	$[AGAT]_{11}$	Puers et al. (1993)
11.1	342 bp	326 bp	Not published	Scherczinger et al. (2000)
11.3	344 bp	328 bp	Not published	STRBase
12	345 bp	329 bp	$[AGAT]_{12}$	Puers et al. (1993)
12.1	346 bp	330 bp	Not published	Budowle & Moretti (1998)
12.2	347 bp	331 bp	Not published	STRBase
12.3	348 bp	332 bp	Not published	STRBase
13	349 bp	333 bp	$[AGAT]_{13}$	Puers et al. (1993)
13.1	350 bp	334 bp	Not published	STRBase
14	353 bp	337 bp	$[AGAT]_{14}$	Puers et al. (1993)
14.1	354 bp	338 bp	Not published	STRBase
15	357 bp	341 bp	$[AGAT]_{15}$	Lazaruk et al. (1998)
15.1	358 bp	342 bp	Not published	STRBase
16	361 bp	345 bp	$[AGAT]_{16}$	Margolis-Nunno et al. (2001)
17	365 bp	349 bp	Not published	Huel et al. (2007)

29 observed alleles

FGA

GenBank Accession: M64982 (allele 21). PCR product sizes of observed alleles

Allele (Repeat #)	Promega PP16	ABI Identifiler	Repeat Structure $[TTTC]_3$ TTT TTCT$[CTTT]_n$ CTCC $[TTCC]_2$	Reference
12.2	308 bp	196 bp	Not published	STRBase
13	310 bp	198 bp	$[TTTC]_3$TTTT TTCT$[CTTT]_5$CTCC$[TTCC]_2$	Jiang et al. (2011)
13.2	312 bp	200 bp	Not published	STRBase
14	314 bp	202 bp	Not published	STRBase
14.3	317 bp	205 bp	Not published	STRBase
15	318 bp	206 bp	$[TTTC]_3$TTTT TTCT$[CTTT]_7$CTCC$[TTCC]_2$	Barber et al. (1996)
15.3	321 bp	209 bp	Not published	STRBase
16	322 bp	210 bp	Not published	STRBase
16.1	323 bp	211 bp	$[TTTC]_3$TTTT TTCT$[CTTT]_5$T$[CTTT]_3$ [CTCC$[TTCC]_2$	Griffiths et al. (1998)
16.2	324 bp	212 bp	Not published	STRBase
17	326 bp	214 bp	$[TTTC]_3$TTTT TTCT$[CTTT]_9$CTCC$[TTCC]_2$	Barber et al. (1996)
17.2	328 bp	216 bp	Not published	STRBase
18	330 bp	218 bp	$[TTTC]_3$TTTT TTCT$[CTTT]_{10}$CTCC$[TTCC]_2$	Barber et al. (1996)
18.2	332 bp	220 bp	$[TTTC]_3$TTTT TT$[CTTT]_{11}$CTCC$[TTCC]_2$	Barber et al. (1996)
19	334 bp	222 bp	$[TTTC]_3$TTTT TTCT$[CTTT]_{11}$CTCC$[TTCC]_2$	Barber et al. (1996)
19.1	335 bp	223 bp	Not published	STRBase
19.2	336 bp	224 bp	$[TTTC]_3$TTTT TT $[CTTT]_{12}$CTCC$[TTCC]_2$	STRBase
19.3	337 bp	225 bp	Not published	STRBase
20	338 bp	226 bp	$[TTTC]_3$TTTT TTCT$[CTTT]_{12}$CTCC$[TTCC]_2$	Barber et al. (1996)
20.1	339 bp	227 bp	Not published	Huel et al. (2007)
20.2	340 bp	228 bp	$[TTTC]_3$TTTT TT $[CTTT]_{13}$CTCC$[TTCC]_2$	Barber et al. (1996)
20.3	341 bp	229 bp	Not published	STRBase
21	342 bp	230 bp	$[TTTC]_3$TTTT TTCT$[CTTT]_{13}$CTCC$[TTCC]_2$	Barber et al. (1996)
21.1	343 bp	231 bp	$[TTTC]_3$TTTT [TTCT$[CTTT]_{10}$**T**$[CTTT]_3$ CTCC$[TTCC]_2$	Allor et al. (2005)
21.2	344 bp	232 bp	$[TTTC]_3$TTTT TT $[CTTT]_{14}$CTCC$[TTCC]_2$	STRBase
21.3	345 bp	233 bp	Not published	STRBase
22	346 bp	234 bp	$[TTTC]_3$TTTT TTCT$[CTTT]_{14}$CTCC$[TTCC]_2$	Barber et al. (1996)
22.1	347 bp	235 bp	Not published	STRBase

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FGA (*Continued*)

Allele (Repeat #)	Promega PP16	ABI Identifiler	Repeat Structure $[TTTC]_3$ TTTT TTCT$[CTTT]_n$ CTCC $[TTCC]_2$	Reference
22.2	348 bp	236 bp	$[TTTC]_3$TTTT TT $[CTTT]_{15}$CTCC$[TTCC]_2$	Barber et al. (1996)
22.3	349 bp	237 bp	Not published	Gill et al. (1996)
23	350 bp	238 bp	$[TTTC]_3$TTTT TTCT$[CTTT]_{15}$CTCC$[TTCC]_2$	Barber et al. (1996)
23.1	351 bp	239 bp	Not published	STRBase
23.2	352 bp	240 bp	$[TTTC]_3$TTTT TT $[CTTT]_{16}$CTCC$[TTCC]_2$	Barber et al. (1996)
23.3	353 bp	241 bp	Not published	STRBase
24	354 bp	242 bp	$[TTTC]_3$TTTT TTCT$[CTTT]_{16}$CTCC$[TTCC]_2$	Barber et al. (1996)
24.1	355 bp	243 bp	Not published	STRBase
24.2	356 bp	244 bp	$[TTTC]_3$TTTT TT $[CTTT]_{17}$CTCC$[TTCC]_2$	Barber et al. (1996)
24.3	357 bp	245 bp	Not published	STRBase
25	358 bp	246 bp	$[TTTC]_3$TTTT TTCT$[CTTT]_{17}$CTCC$[TTCC]_2$	Barber et al. (1996)
25.1	359 bp	247 bp	Not published	STRBase
25.2	360 bp	248 bp	$[TTTC]_3$TTTT TT $[CTTT]_{18}$CTCC$[TTCC]_2$	STRBase
25.3	361 bp	249 bp	Not published	STRBase
26	362 bp	250 bp	$[TTTC]_3$TTTT TTCT$[CTTT]_{18}$CTCC$[TTCC]_2$	Barber et al. (1996)
26.1	363 bp	251 bp	Not published	STRBase
26.2	364 bp	252 bp	Not published	Huel et al. (2007)
26.3	365 bp	253 bp	Not published	STRBase
27 (a)	366 bp	254 bp	$[TTTC]_3$TTTT TTCT$[CTTT]_{19}$CTCC$[TTCC]_2$	Barber et al. (1996)
27 (b)	366 bp	254 bp	$[TTTC]_3$TTTT TTCT$[CTTT]_{13}$CCTT$[CTTT]_5$CTCC $[TTCC]_2$	Griffiths et al. (1998)
27.1	367 bp	255 bp	Not published	STRBase
27.2	368 bp	256 bp	Not published	Huel et al. (2007)
27.3	369 bp	257 bp	Not published	STRBase
28	370 bp	258 bp	$[TTTC]_3$TTTT TTCT$[CTTT]_{20}$CTCC$[TTCC]_2$	Barber et al. (1996)
28.2	372 bp	260 bp	$[TTTC]_3$TTTT TT $[CTTT]_{21}$CTCC$[TTCC]_2$	STRBase
29	374 bp	262 bp	$[TTTC]_3$TTTT TTCT$[CTTT]_{15}$CCTT$[CTTT]_5$CTCC $[TTCC]_2$	Barber et al. (1996)
29.1	375 bp	263 bp	Not published	STRBase
29.2	376 bp	264 bp	Not published	STRBase
30	378 bp	266 bp	$[TTTC]_3$TTTT TTCT$[CTTT]_{16}$CCTT$[CTTT]_5$CTCC $[TTCC]_2$	Griffiths et al. (1998)

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FGA (*Continued*)

Allele (Repeat #)	Promega PP16	ABI Identifiler	Repeat Structure $[TTTC]_3$ TTTT TTCT$[CTTT]_n$ CTCC $[TTCC]_2$	Reference
30.2	380 bp	268 bp	$[TTTC]_4$TTTT TT $[CTTT]_{14}[CTTC]_3[CTTT]_3$CTCC $[TTCC]_4$	Barber et al. (1996)
31	382 bp	270 bp	Not published	STRBase
31.2	384 bp	272 bp	$[TTTC]_4$TTTT TT $[CTTT]_{15}[CTTC]_3[CTTT]_3$CTCC $[TTCC]_4$	Griffiths et al. (1998)
32	386 bp	274 bp	Not published	SGM Plus
32.1	387 bp	275 bp	Not published	STRBase
32.2	388 bp	276 bp	$[TTTC]_4$TTTT TT $[CTTT]_{16}[CTTC]_3[CTTT]_3$CTCC $[TTCC]_4$	Griffiths et al. (1998)
33.1	391 bp	279 bp	$[TTTC]_3$TTTTTTCT$[CTTT]_{13}$TTTCT$[CTTT]_{11}$CTC C$[TTCC]_2$	Allor et al. (2005)
33.2	392 bp	280 bp	$[TTTC]_4$TTTT TT $[CTTT]_{17}[CTTC]_3[CTTT]_3$CTCC $[TTCC]_4$	Griffiths et al. (1998)
34.1	395 bp	283 bp	$[TTTC]_3$TTTTTTCT$[CTTT]_{13}$TTTCT$[CTTT]_{12}$CTC C$[TTCC]_2$	Allor et al. (2005)
34.2	396 bp	284 bp	$[TTTC]_4$TTTT TT $[CTTT]_{18}[CTTC]_3[CTTT]_3$CTCC $[TTCC]_4$	Barber et al. (1996)
35.2	400 bp	288 bp	Not published	STRBase
41.1	423 bp	311 bp	Not published	STRBase
41.2	424 bp	312 bp	$[TTTC]_4$TTTT TT $[CTTT]_{11}[CTGT]_3[CTTT]_{11}$ $[CTTC]_3$ $[CTTT]_3$CTCC$[TTCC]_4$	Allor et al. (2005)
42	426 bp	314 bp	Not published	STRBase
42.1	427 bp	315 bp	Not published	STRBase
42.2	428 bp	316 bp	$[TTTC]_4$TTTT TT $[CTTT]_8$ $[CTGT]_4[CTTT]_{13}$ $[CTTC]_3[CTTT]_3$CTCC$[TTCC]_4$	Griffiths et al. (1998)
43.1	431 bp	319 bp	Not published	STRBase
43.2	432 bp	320 bp	$[TTTC]_4$TTTT TT $[CTTT]_8$ $[CTGT]_5[CTTT]_{13}$ $[CTTC]_4[CTTT]_3$CTCC$[TTCC]_4$	Griffiths et al. (1998)
44	434 bp	322 bp	Not published	Steinlechner et al. (2002)
44.2	436 bp	324 bp	$[TTTC]_4$TTTT TT $[CTTT]_{11}[CTGT]_3[CTTT]_{14}$ $[CTTC]_3[CTTT]_3$CTCC$[TTCC]_4$	Griffiths et al. (1998)
44.3	437 bp	325 bp	Not published	STRBase
45	438 bp	326 bp	Not published	STRBase
45.1	439 bp	327 bp	Not published	STRBase

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FGA (*Continued*)

Allele (Repeat #)	Promega PP16	ABI Identifiler	Repeat Structure $[TTTC]_3$ TTTT TTCT$[CTTT]_n$ CTCC $[TTCC]_2$	Reference
45.2	440 bp	328 bp	$[TTTC]_4$TTTT TT $[CTTT]_{10}[CTGT]_5[CTTT]_{13}$ $[CTTC]_4[CTTT]_3$CTCC$[TTCC]_4$	Griffiths et al. (1998)
46	442 bp	330 bp	Not published	STRBase
46.1	443 bp	331 bp	Not published	STRBase
46.2	444 bp	332 bp	$[TTTC]_4$TTTT TT $[CTTT]_{12}[CTGT]_5[CTTT]_{13}$ $[CTTC]_3[CTTT]_3$CTCC$[TTCC]_4$	Barber et al. (1996)
47	446 bp	334 bp	Not published	STRBase
47.2	448 bp	336 bp	$[TTTC]_4$TTTT TT $[CTTT]_{12}[CTGT]_5[CTTT]_{14}$ $[CTTC]_3[CTTT]_3$CTCC$[TTCC]_4$	Griffiths et al. (1998)
48	450 bp	338 bp	Not published	STRBase
48.2	452 bp	340 bp	$[TTTC]_4$TTTT TT $[CTTT]_{14}[CTGT]_3[CTTT]_{14}$ $[CTTC]_4[CTTT]_3$CTCC$[TTCC]_4$	Griffiths et al. (1998)
49	454 bp	342 bp	Not published	STRBase
49.1	455 bp	343 bp	Not published	STRBase
49.2	456 bp	344 bp	Not published	SGM Plus
50.2	460 bp	344 bp	$[TTTC]_4$TTTT TT $[CTTT]_{14}[CTGT]_4[CTTT]_{15}$ $[CTTC]_4[CTTT]_3$CTCC$[TTCC]_4$	Griffiths et al. (1998)
50.3	461 bp	345 bp	Not published	STRBase
51	462 bp	346 bp	Not published	STRBase
51.2	464 bp	348 bp	Not published	SGM Plus

95 observed alleles

TH01

GenBank Accession: D00269 (allele 9). PCR product sizes of observed alleles

Allele (Repeat #)	Promega PowerPlex 16	ABI Identifiler	Repeat Structure $[AATG]_n$ *or other strand* $[TCAT]_n$	Reference
3	152 bp	160 bp	$[AATG]_3$	Espinheira et al. (1996)
4	156 bp	164 bp	$[AATG]_4$	Griffiths et al. (1998)
5	160 bp	168 bp	$[AATG]_5$	Brinkmann et al. (1996b)
5.3	163 bp	171 bp	Not published	SGM Plus manual
6	164 bp	172 bp	$[AATG]_6$	Brinkmann et al. (1996b)
6.1	165 bp	173 bp	Not published	STRBase
6.3	167 bp	175 bp	$[AATG]_3ATG[AATG]_3$	Klintschar et al. (1998)
7	168 bp	176 bp	$[AATG]_7$	Brinkmann et al. (1996b)
7.1	169 bp	177 bp	Not published	STRBase
7.3	171 bp	179 bp	Not published	STRBase
8	172 bp	180 bp	$[AATG]_8$	Brinkmann et al. (1996b)
8.1	173 bp	181 bp	Not published	STRBase
8.3	175 bp	183 bp	$[AATG]_5ATG[AATG]_3$	Brinkmann et al. (1996b)
9	176 bp	184 bp	$[AATG]_9$	Brinkmann et al. (1996b)
9.1	177 bp	185 bp	Not published	STRBase
9.2	178 bp	186 bp	Not published	STRBase
9.3	179 bp	187 bp	$[AATG]_6ATG[AATG]_3$	Brinkmann et al. (1996b)
10	180 bp	188 bp	$[AATG]_{10}$	Brinkmann et al. (1996b)
10.3	183 bp	191 bp	$[AATG]_6ATG[AATG]_4$	Brinkmann et al. (1996b)
11	184 bp	192 bp	$[AATG]_{11}$	Brinkmann et al. (1996b)
12	188 bp	196 bp	$[AATG]_{12}$	van Oorschot et al. (1994)
13	192 bp	200 bp	Not published	STRBase
13.3	195 bp	203 bp	$[AATG][AACG][AATG]_8$ $ATG[AATG]_3$	Gene et al. (1996); Griffiths et al. (1998)
14	196 bp	204 bp	Not published	SGM Plus manual

24 observed alleles

TPOX

GenBank Accession: M68651 (allele 11). PCR product sizes of observed alleles

Allele (Repeat #)	Promega PP 16	ABI Identifiler	Repeat Structure $[AATG]_n$	Reference
4	254 bp	209 bp	$[AATG]_4$	STRBase
5	258 bp	213 bp	$[AATG]_5$	STRBase
6	262 bp	217 bp	$[AATG]_6$	Lazaruk et al. (1998)
7	266 bp	221 bp	$[AATG]_7$	Amorim et al. (1996)
7.1	267 bp	222 bp	Not published	STRBase
7.2	268 bp	223 bp	Not published	STRBase
7.3 or **8(D48Adel)**	269 bp	224 bp	$[AATG]_8$ with A deletion 48 bp downstream	Allor et al. (2005)
8	270 bp	225 bp	$[AATG]_8$	Puers et al. (1993)
9	274 bp	229 bp	$[AATG]_9$	Puers et al. (1993)
10	278 bp	233 bp	$[AATG]_{10}$	Puers et al. (1993)
10.1	279 bp	234 bp	Not published	STRBase
10.3 or **11(D157Gdel)**	281 bp	236 bp	$[AATG]_{11}$ with G deletion 157 bp downstream	Kline et al. (2010)
11	282 bp	237 bp	$[AATG]_{11}$	Puers et al. (1993)
12	286 bp	241 bp	$[AATG]_{12}$	Puers et al. (1993)
13	290 bp	245 bp	$[AATG]_{13}$	Amorim et al. (1996)
13.1	291 bp	246 bp	Not published	STRBase
14	294 bp	249 bp	$[AATG]_{14}$	Huang et al. (1995)
15	298 bp	253 bp	Not published	STRBase
16	302 bp	257 bp	Not published	STRBase

19 observed alleles

VWA

GenBank Accession: M25858 (allele 18). PCR product sizes of observed alleles

Allele (Repeat #)	Promega PP 16	ABI Identifiler	Repeat Structure TCTA[TCTG]$_{3-4}$[TCTA]$_n$	Reference
10	123 bp	152 bp	TCTA TCTG TCTA [TCTG]$_4$[TCTA]$_3$	Griffiths et al. (1998)
11	127 bp	156 bp	TCTA[TCTG]$_3$[TCTA]$_7$	Brinkmann et al. (1996b)
12	131 bp	160 bp	TCTA[TCTG]$_4$[TCTA]$_7$	Griffiths et al. (1998)
13 (a)	135 bp	164 bp	[TCTA]$_2$[TCTG]$_4$[TCTA]$_3$TCCA [TCTA]$_3$	Griffiths et al. (1998)
13 (b)	135 bp	164 bp	TCTA[TCTA]$_4$[TCTA]$_8$TC**C**ATCTA	Brinkmann et al. (1996b)
13 (c)	135 bp	164 bp	TCTA[TCTA]$_4$[TCTA]$_8$TC**T**ATCTA	Brinkmann et al. (1996b)
13 (d)	135 bp	164 bp	TCTA[TCTA]$_3$[TCTA]$_9$	Cruz et al. (2004)
14 (a)	139 bp	168 bp	TCTA[TCTG]$_4$[TCTA]$_9$	Brinkmann et al. (1996b)
14 (b)	139 bp	168 bp	TCTA TCTG TCTA[TCTG]$_4$[TCTA]$_3$ TCCA[TCTA]$_3$	Brinkmann et al. (1996b)
14 (c)	139 bp	168 bp	TCTA [TCTG]$_5$[TCTA]$_3$TCCA[TCTA]$_3$	Lins et al. (1998)
14 (d)	139 bp	168 bp	TCTA[TCTA]$_3$[TCTA]$_{10}$	Cruz et al. (2004)
15 (a)	143 bp	172 bp	TCTA[TCTG]$_4$[TCTA]$_{10}$	Brinkmann et al. (1996b)
15 (b)	143 bp	172 bp	TCTA[TCTG]$_3$[TCTA]$_{11}$	Brinkmann et al. (1996b)
15.2	145 bp	174 bp	[TCTA]$_2$[TCTG]$_4$[TCTA]$_5$T–A[TCTA]$_4$	Gill et al. (1995)
16 (a)	147 bp	176 bp	TCTA[TCTG]$_4$[TCTA]$_{11}$	Brinkmann et al. (1996b)
16 (b)	147 bp	176 bp	TCTA[TCTG]$_3$[TCTA]$_{12}$	Brinkmann et al. (1996b)
16.1	148 bp	177 bp	Not published	STRBase
17	151 bp	180 bp	TCTA[TCTG]$_4$[TCTA]$_{12}$	Brinkmann et al. (1996b)
18 (a)	155 bp	184 bp	TCTA[TCTG]$_4$[TCTA]$_{13}$	Brinkmann et al. (1996b)
18 (b)	155 bp	184 bp	TCTA[TCTG]$_5$[TCTA]$_{12}$	Brinkmann et al. (1996b)
18.1	156 bp	185 bp	TCTA[TCTG]$_4$[TCTA]$_{12}$A(TCTA)	Kido et al. (2003)
18.2	157 bp	186 bp	Not published	SGM Plus
18.3	158 bp	187 bp	TCTA[TCTG]$_4$[TCTA]$_{11}$TCA[TCTA]$_2$	Dauber et al. (2008)
19 (a)	159 bp	188 bp	TCTA[TCTG]$_4$[TCTA]$_{14}$	Brinkmann et al. (1996b)
19 (b)	159 bp	188 bp	TCTA[TCTG]$_5$[TCTA]$_{13}$	Cruz et al. (2004)
19.2	161 bp	190 bp	Not published	SGM Plus
20 (a)	163 bp	192 bp	TCTA[TCTG]$_4$[TCTA]$_{15}$	Brinkmann et al. (1996b)
20 (b)	163 bp	192 bp	TCTA[TCTG]$_3$[TCTA]$_{16}$	Cruz et al. (2004)

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VWA (*Continued*)

Allele (Repeat #)	Promega PP 16	ABI Identifiler	Repeat Structure $TCTA[TCTG]_{3–4}[TCTA]_n$	Reference
21 (a)	167 bp	196 bp	$TCTA[TCTG]_4[TCTA]_{16}$	Brinkmann et al. (1996b)
21 (b)	167 bp	196 bp	$TCTA[TCTG]_5[TCTA]_{15}$	Cruz et al. (2004)
21 (c)	167 bp	196 bp	$TCTA[TCTG]_6[TCTA]_{14}$	Cruz et al. (2004)
22 (a)	171 bp	200 bp	$TCTA[TCTG]_4[TCTA]_{17}$	Brinkmann et al. (1996b)
22 (b)	171 bp	200 bp	$TCTA[TCTG]_5[TCTA]_{16}$	Cruz et al. (2004)
23	175 bp	204 bp	Not published	SGM Plus
24	179 bp	208 bp	Not published	STRBase
25	183 bp	212 bp	Not published	STRBase

36 observed alleles

D3S1358

GenBank Accession: AC099539 (allele 16). PCR product sizes of observed alleles

Allele (Repeat #)	Promega PowerPlex 16	ABI Identifiler	Repeat Structure $TCTA[TCTG]_{2-3}[TCTA]_n$	Reference
6	91 bp	89 bp	Not published	Phillips et al. (2011)
8	99 bp	97 bp	Not published	STRBase
8.3	102 bp	100 bp	Not published	STRBase
9	103 bp	101 bp	Not published	Allor et al. (2005)
10	107 bp	105 bp	Not published	STRBase
11 (a)	111 bp	109 bp	$TCTA[TCTG]_3[TCTA]_7$	Heinrich et al. (2005)
11 (b)	111 bp	109 bp	$TCTA[TCTG]_2[TCTA]_8$	Dauber et al. (2009)
12	115 bp	113 bp	$TCTA[TCTG]_3[TCTA]_8$	Heinrich et al. (2005)
13	119 bp	117 bp	$TCTA[TCTG]_2[TCTA]_{10}$	Mornhinweg et al. (1998)
13.3	122 bp	120 bp	Not published	STRBase
14	123 bp	121 bp	$TCTA[TCTG]_2[TCTA]_{11}$	Szibor et al. (1998)
14.2	125 bp	123 bp	Not published	STRBase
14.3	126 bp	124 bp	Not published	STRBase
15 (a)	127 bp	125 bp	$TCTA[TCTG]_3[TCTA]_{11}$	Szibor et al. (1998)
15 (b)	127 bp	125 bp	$TCTA[TCTG]_2[TCTA]_{12}$	Szibor et al. (1998)
15 (c)	127 bp	125 bp	$TCTA[TCTG]_1[TCTA]_{13}$	Heinrich et al. (2005)
15.1	128 bp	126 bp	$TCTA[TCTG]A[TCTA]_{13}$	Allor et al. (2005)
15.2	129 bp	127 bp	$TCTA[TCTG]_3TC\ [TCTA]_{11}$	Heinrich et al. (2005)
15.3	130 bp	128 bp	Not published	STRBase
16 (a)	131 bp	129 bp	$TCTA[TCTG]_3[TCTA]_{12}$	Szibor et al. (1998)
16 (b)	131 bp	129 bp	$TCTA[TCTG]_2[TCTA]_{13}$	Mornhinweg et al. (1998)
16 (c)	131 bp	129 bp	$TCTA[TCTG]_1[TCTA]_{14}$	Heinrich et al. (2005)
16.2	133 bp	131 bp	Not published	Budowle et al. (1997)
17 (a)	135 bp	133 bp	$TCTA[TCTG]_3[TCTA]_{13}$	Szibor et al. (1998)
17 (b)	135 bp	133 bp	$TCTA[TCTG]_2[TCTA]_{14}$	Mornhinweg et al. (1998)
17 (c)	135 bp	133 bp	$TCTA[TCTG]_1[TCTA]_{15}$	Heinrich et al. (2005)
17.1	136 bp	134 bp	Not published	Allor et al. (2005)
17.2	137 bp	135 bp	Not published	STRBase

(Continued)

D3S1358 (*Continued*)

Allele (Repeat #)	Promega PowerPlex 16	ABI Identifiler	Repeat Structure $TCTA[TCTG]_{2-3}[TCTA]_n$	Reference
18 (a)	139 bp	137 bp	$TCTA[TCTG]_4[TCTA]_{13}$	Heinrich et al. (2005)
18 (b)	139 bp	137 bp	$TCTA[TCTG]_3[TCTA]_{14}$	Szibor et al. (1998)
18 (c)	139 bp	137 bp	$TCTA[TCTG]_2[TCTA]_{15}$	Heinrich et al. (2005)
18.1	140 bp	138 bp	Not published	STRBase
18.2	141 bp	139 bp	Not published	STRBase
18.3	142 bp	140 bp	Not published	STRBase
19 (a)	143 bp	141 bp	$TCTA[TCTG]_3[TCTA]_{15}$	Mornhinweg et al. (1998)
19 (b)	143 bp	141 bp	$TCTA[TCTG]_2[TCTA]_{16}$	Heinrich et al. (2005)
20 (a)	147 bp	145 bp	$TCTA[TCTG]_3[TCTA]_{16}$	Mornhinweg et al. (1998)
20 (b)	147 bp	145 bp	$TCTA[TCTG]_2[TCTA]_{17}$	Heinrich et al. (2005)
20.1	148 bp	146 bp	Not published	Allor et al. (2005)
21	151 bp	149 bp	$TCTA[TCTG]_3[TCTA]_{17}$	Heinrich et al. (2005)
21.1	152 bp	150 bp	Not published	STRBase
22	155 bp	153 bp	$TCTA[TCTG]_3[TCTA]_4TCTG[TCTA]_{13}$	A. Raziel (unpublished)
23	159 bp	157 bp	$TCTA[TCTG]_3[TCTA]_{19}$	Kline et al. (2010)
26	171 bp	169 bp	$TCTA[TCTG]_3[TCTA]_9[TCTG]_2[TCTA]_{11}$	Grubwieser et al. (2005)

44 observed alleles

D5S818

GenBank Accession: AC008512 (allele 11). PCR product sizes of observed alleles

Allele (Repeat #)	Promega PowerPlex 16	ABI Identifiler	Repeat Structure $[AGAT]_n$	Reference
4	107 bp	122 bp	Not published	STRBase; Cho et al. (2010)
5	111 bp	126 bp	Not published	STRBase
6	115 bp	130 bp	Not published	STRBase; Cho et al. (2010)
7	119 bp	134 bp	$[AGAT]_7$	Lins et al. (1998)
8	123 bp	138 bp	$[AGAT]_8$	Lins et al. (1998)
9	127 bp	142 bp	$[AGAT]_9$	Lins et al. (1998)
10	131 bp	146 bp	$[AGAT]_{10}$	Lins et al. (1998)
10.1	132 bp	147 bp	A $[AGAT]_{10}$	Kline et al. (2010)
11	135 bp	150 bp	$[AGAT]_{11}$	Lins et al. (1998)
11 (D55G→T)	135 bp	150 bp	$[AGAT]_{11}$ with G→T 55 bp downstream	Jiang et al. (2011)
11.1	136 bp	151 bp	Not published	STRBase
11.3	138 bp	153 bp	Not published	STRBase
12	139 bp	154 bp	$[AGAT]_{12}$	Lins et al. (1998)
12.1	140 bp	155 bp	Not published	STRBase
12.3	142 bp	157 bp	$[AGAT]_3$ GAT $[AGAT]_9$	Allor et al. (2005)
13	143 bp	158 bp	$[AGAT]_{13}$	Lins et al. (1998)
13.3	146 bp	161 bp	Not published	STRBase
14	147 bp	162 bp	$[AGAT]_{14}$	Lins et al. (1998)
15	151 bp	166 bp	$[AGAT]_{15}$	Lins et al. (1998)
16	155 bp	170 bp	$[AGAT]_{16}$	Profiler Plus
17	159 bp	174 bp	Not published	STRBase
18	163 bp	178 bp	$[AGAT]_{14}$ ACAT $[AGAT]_3$	Allor et al. (2005)
20	171 bp	186 bp	Not published	STRBase
29	207 bp	222 bp	$[AGAT]_{12}$ {48 bp} $[AGAT]_5$	Kline et al. (2010)

24 observed alleles

D7S820

GenBank Accession: AC004848 (allele 13). PCR product sizes of observed alleles

Allele (Repeat #)	Promega PowerPlex 16	ABI Identifiler	Repeat Structure $[GATA]_n$	Reference
5	211 bp	253 bp	Not published	STRBase; Cho et al. (2010)
5.2 or **9(D41-14 bp-del)**	213 bp	255 bp	$[GATA]_9$ with 14 bp deletion 41 bp downstream	Allor et al. (2005)
6	215 bp	257 bp	$[GATA]_6$	Lins et al. (1998)
6.2	217 bp	259 bp	Not published	STRBase
6.3 or 7(D14Tdel)	218 bp	260 bp	$[GATA]_7$ with T deletion 14 bp downstream	Kline et al. (2010)
7	219 bp	261 bp	$[GATA]_7$	Lins et al. (1998)
7.1	220 bp	262 bp	Not published	STRBase
7.3	222 bp	264 bp	Not published	Ayres et al. (2002)
8	223 bp	265 bp	$[GATA]_8$	Lins et al. (1998)
8.1	224 bp	266 bp	Not published	Balamurugan et al. (2000)
8.2	225 bp	267 bp	Not published	STRBase
8.3	226 bp	268 bp	$[GATA]_9$ with A deletion 22 bp downstream	Kline et al. (2010)
9	227 bp	269 bp	$[GATA]_9$	Lins et al. (1998)
9.1	228 bp	270 bp	Not published	Ayres et al. (2002)
9.2	229 bp	271 bp	Not published	STRBase
9.3	230 bp	272 bp	Not published	Huel et al. (2007)
10	231 bp	273 bp	$[GATA]_{10}$	Lins et al. (1998)
10.1 or **10(U1Ains)**	232 bp	274 bp	A $[GATA]_{10}$	Kline et al. (2010)
10.3	234 bp	276 bp	Not published	Allor et al. (2005)
11	235 bp	277 bp	$[GATA]_{11}$	Lins et al. (1998)
11.1	236 bp	278 bp	Not published	Huel et al. (2007)
11.2	237 bp	279 bp	Not published	STRBase
11.3	238 bp	280 bp	Not published	Allor et al. (2005)
12	239 bp	281 bp	$[GATA]_{12}$	Lins et al. (1998)
12.1	240 bp	282 bp	Not published	Allor et al. (2005)
12.2	241 bp	283 bp	Not published	STRBase
12.3	242 bp	284 bp	Not published	STRBase

(Continued)

D7S820 (*Continued*)

Allele (Repeat #)	Promega PowerPlex 16	ABI Identifiler	Repeat Structure $[GATA]_n$	Reference
13	243 bp	285 bp	$[GATA]_{13}$	Lins et al. (1998)
13.1	244 bp	286 bp	Not published	Allor et al. (2005)
14	247 bp	289 bp	$[GATA]_{14}$	Lins et al. (1998)
14.1	248 bp	290 bp	Not published	STRBase
15	251 bp	293 bp	Not published	Huel et al. (2007)
16	255 bp	297 bp	Not published	Ayres et al. (2002)

33 observed alleles

D8S1179 (listed as D6S502 in early papers)
GenBank Accession: AF216671 (allele 13). PCR product sizes of observed alleles

Allele (Repeat #)	Promega PowerPlex 16	ABI Identifiler	Repeat Structure $[TCTR]_n$	Reference
6	199 bp	119 bp	Not published	STRBase
7	203 bp	123 bp	$[TCTA]_7$	Griffiths et al. (1998)
8	207 bp	127 bp	$[TCTA]_8$	Barber & Parkin (1996)
9	211 bp	131 bp	$[TCTA]_9$	Barber & Parkin (1996)
10	215 bp	135 bp	$[TCTA]_{10}$	Barber & Parkin (1996)
10.1	216 bp	136 bp	Not published	STRBase
10.2	217 bp	137 bp	Not published	STRBase
11	219 bp	139 bp	$[TCTA]_{11}$	Barber & Parkin (1996)
12	223 bp	143 bp	$[TCTA]_{12}$	Barber & Parkin (1996)
12.1	224 bp	144 bp	Not published	STRBase
12.2	225 bp	145 bp	Not published	STRBase
12.3	226 bp	146 bp	Not published	STRBase
13 (a)	227 bp	147 bp	$[TCTA]_1[TCTG]_1[TCTA]_{11}$	Barber & Parkin (1996)
13 (b)	227 bp	147 bp	$[TCTA]_2[TCTG]_1[TCTA]_{10}$	Kline et al. (2010)
13 (c)	227 bp	147 bp	$[TCTA]_1[TCTG]_1$TGTA $[TCTA]_{10}$	Kline et al. (2010)
13 (d)	227 bp	147 bp	$[TCTA]_{13}$	Kline et al. (2010)
13.1	228 bp	148 bp	Not published	STRBase
13.2	229 bp	149 bp	Not published	STRBase
13.3	230 bp	150 bp	Not published	STRBase
14	231 bp	151 bp	$[TCTA]_2[TCTG]_1[TCTA]_{11}$	Barber & Parkin (1996)
14.1	232 bp	152 bp	Not published	STRBase
14.2	233 bp	153 bp	Not published	STRBase
15	235 bp	155 bp	$[TCTA]_2[TCTG]_1[TCTA]_{12}$	Barber & Parkin (1996)
15.1	236 bp	156 bp	Not published	STRBase
15.2	237 bp	157 bp	Not published	STRBase
15.3	238 bp	158 bp	Not published	STRBase
16	239 bp	159 bp	$[TCTA]_2[TCTG]_1[TCTA]_{13}$	Barber & Parkin (1996)
16.1	240 bp	160 bp	Not published	STRBase
17	243 bp	163 bp	$[TCTA]_2[TCTG]_2[TCTA]_{13}$	Barber & Parkin (1996)

(Continued)

D8S1179 (*Continued*)

Allele (Repeat #)	Promega PowerPlex 16	ABI Identifiler	Repeat Structure $[TCTR]_n$	Reference
17.1	244 bp	164 bp	Not published	STRBase
17.2	245 bp	165 bp	Not published	STRBase
18	247 bp	167 bp	$[TCTA]_2[TCTG]_1[TCTA]_{15}$	Barber & Parkin (1996)
19	251 bp	171 bp	$[TCTA]_2[TCTG]_2[TCTA]_{15}$	Griffiths et al. (1998)
20	255 bp	175 bp	Not published	STRBase

34 observed alleles

D13S317

GenBank Accession: AL353628 (allele 11). PCR product sizes of observed alleles

Allele (Repeat #)	Promega PowerPlex 16	ABI Identifiler*	Repeat Structure $[TATC]_n$	Reference
5	157 bp	193 bp	Not published	Allor et al. (2005)
6	161 bp	197 bp	Not published	Allor et al. (2005)
7	165 bp	201 bp	$[TATC]_7$	Lins et al. (1998)
7.1	166 bp	202 bp	Not published	Allor et al. (2005)
8	169 bp	205 bp	$[TATC]_8$	Lins et al. (1998)
8.1	170 bp	206 bp	Not published	STRBase
9	173 bp	209 bp	$[TATC]_9$	Lins et al. (1998)
9.1	174 bp	210 bp	Not published	STRBase
10 (a)	177 bp	213 bp	$[TATC]_{10}$	Lins et al. (1998)
10 (b)	177 bp	213 bp	$[TATC]_{10}$ AATC	Lins et al. (1998)
10.3	180 bp	216 bp	Not published	STRBase
11	181 bp	217 bp	$[TATC]_{11}$	Lins et al. (1998)
11.1	182 bp	218 bp	Not published	STRBase
11.3	184 bp	220 bp	Not published	STRBase
12	185 bp	221 bp	$[TATC]_{12}$	Lins et al. (1998)
13	189 bp	225 bp	$[TATC]_{13}$	Lins et al. (1998)
13.3	192 bp	228 bp	Not published	STRBase
14	193 bp	229 bp	$[TATC]_{14}$	Lins et al. (1998)
14.3	196 bp	232 bp	Not published	STRBase
15	197 bp	233 bp	$[TATC]_{15}$	Lins et al. (1998)
16	201 bp	237 bp	Not published	Huel et al. (2007)
17	205 bp	241 bp	Not published	Huel et al. (2007)

22 observed alleles

**Mobility modified PCR products so observed size will not equal expected DNA sequence size*

D16S539

GenBank Accession: AC024591 (allele 11). PCR product sizes of observed alleles

Allele (Repeat #)	Promega PowerPlex 16	ABI Identifiler*	Repeat Structure $[GATA]_n$	Reference
4	260 bp	248 bp	Not published	STRBase
5	264 bp	252 bp	$[GATA]_5$	Lins et al. (1998)
6	268 bp	256 bp	$[GATA]_6$	STRBase
7	272 bp	260 bp	$[GATA]_7$	STRBase
8	276 bp	264 bp	$[GATA]_8$	Lins et al. (1998)
8.3	279 bp	267 bp	Not published	STRBase
9	280 bp	268 bp	$[GATA]_9$	Lins et al. (1998)
9.3	283 bp	271 bp	Not published	STRBase
10	284 bp	272 bp	$[GATA]_{10}$	Lins et al. (1998)
10.1	285 bp	273 bp	Not published	STRBase
10.3	287 bp	275 bp	Not published	STRBase
11	288 bp	276 bp	$[GATA]_{11}$	Lins et al. (1998)
11.1	289 bp	277 bp	Not published	STRBase
11.3	291 bp	279 bp	Not published	STRBase
12	292 bp	280 bp	$[GATA]_{12}$	Lins et al. (1998)
12.1	293 bp	281 bp	Not published	Huel et al. (2007)
12.2	294 bp	282 bp	Not published	STRBase
12.3	295 bp	283 bp	Not published	STRBase
13	296 bp	284 bp	$[GATA]_{13}$	Lins et al. (1998)
13.1	297 bp	285 bp	Not published	STRBase
13.2	298 bp	286 bp	Not published	STRBase
13.3	299 bp	287 bp	Not published	STRBase
14	300 bp	288 bp	$[GATA]_{14}$	Lins et al. (1998)
14.1	301 bp	289 bp	Not published	STRBase
14.2	302 bp	290 bp	Not published	STRBase
14.3	303 bp	291 bp	Not published	STRBase
15	304 bp	292 bp	$[GATA]_{15}$	Lins et al. (1998)
16	308 bp	296 bp	Not published	Huel et al. (2007)
17	312 bp	300 bp	Not published	STRBase

29 observed alleles

**Mobility modified PCR products so observed size will not equal expected DNA sequence size*

D18S51

GenBank Accession: AP001534 (allele 18). PCR product sizes of observed alleles

Allele (Repeat #)	Promega PP 16	ABI Identifiler	Repeat Structure $[AGAA]_n$ 3′ flanking region	Reference
5.3 or **8(D1–9 bp-del)**	281 bp	259 bp	$[AGAA]_8$ immediate downstream deletion of (AAAGAGAGA)	Kline et al. (2010)
6	282 bp	260 bp	Not published	STRBase
7	286 bp	264 bp	Not published	Allor et al. (2005)
8	290 bp	268 bp	$[AGAA]_8$	Griffiths et al. (1998)
9	294 bp	272 bp	$[AGAA]_9$	Barber & Parkin (1996)
9.2*	296 bp	274 bp	Not published	STRBase
10	298 bp	276 bp	$[AGAA]_{10}$	Barber & Parkin (1996)
10.2	300 bp	278 bp	Not published	SGM Plus
11	302 bp	280 bp	$[AGAA]_{11}$	Barber & Parkin (1996)
11.1**	303 bp	281 bp	Not published	STRBase
11.2	304 bp	282 bp	$[AGAA]_{12}$ with AG deletion in 3′ flanking region	Dauber et al. (2008)
11.3	305 bp	283 bp	Not published	STRBase
12	306 bp	284 bp	$[AGAA]_{12}$	Barber & Parkin (1996)
12.2	308 bp	286 bp	Not published	Allor et al. (2005)
12.3	309 bp	287 bp	Not published	STRBase
13	310 bp	288 bp	$[AGAA]_{13}$	Barber & Parkin (1996)
13.1	311 bp	289 bp	Not published	STRBase; Phillips et al. (2011)
13.2	312 bp	290 bp	$[AGAA]_{13}$ [AGAG]AG	Barber & Parkin (1996)
13.3 or **14(D76Adel)**	313 bp	291 bp	$[AGAA]_{14}$ with A deletion 76 bp downstream	Allor et al. (2005)
14	314 bp	292 bp	$[AGAA]_{14}$	Barber & Parkin (1996)
14.1	315 bp	293 bp	Not published	STRBase
14.2	316 bp	294 bp	$[AGAA]_{14}$ [AGAG]AG	Barber & Parkin (1996)
15	318 bp	296 bp	$[AGAA]_{15}$	Barber & Parkin (1996)
15.1	319 bp	297 bp	Not published	STRBase
15.2	320 bp	298 bp	$[AGAA]_{15}$ [AGAG]AG	Barber & Parkin (1996)
15.3	321 bp	299 bp	Not published	STRBase
16	322 bp	300 bp	$[AGAA]_{16}$	Barber & Parkin (1996)
16.1	323 bp	301 bp	$[AGAA]_3$**A**$[AGAA]_{13}$	Allor et al. (2005)

(Continued)

D18S51 (*Continued*)

Allele (Repeat #)	Promega PP 16	ABI Identifiler	Repeat Structure $[AGAA]_n$ 3′ flanking region	Reference
16.2	324 bp	302 bp	$[AGAA]_{16}$ [AGAG]AG	Dauber et al. (2009)
16.3	325 bp	303 bp	Not published	STRBase
17	326 bp	304 bp	$[AGAA]_{17}$	Barber & Parkin (1996)
17.1	327 bp	305 bp	Not published	SGM Plus
17.2	328 bp	306 bp	$[AGAA]_{17}$ [AGAG]AG	Gill et al. (1996)
17.3	329 bp	307 bp	Not published	Allor et al. (2005)
18	330 bp	308 bp	$[AGAA]_{18}$	Barber & Parkin (1996)
18.1	331 bp	309 bp	Not published	Allor et al. (2005)
18.2	332 bp	310 bp	Not published	STRBase
18.3	333 bp	311 bp	Not published	STRBase
19	334 bp	312 bp	$[AGAA]_{19}$	Barber & Parkin (1996)
19.1	335 bp	313 bp	Not published	STRBase
19.2	336 bp	314 bp	$[AGAA]_{19}$ [AGAG]AG	Gill et al. (1996)
20	338 bp	316 bp	$[AGAA]_{20}$	Barber & Parkin (1996)
20.1	339 bp	317 bp	$[AGAA]_{20}$ AAAAG	Dauber et al. (2009)
20.2	340 bp	318 bp	Not published	Allor et al. (2005)
21	342 bp	320 bp	$[AGAA]_{21}$	Barber & Parkin (1996)
21.1	343 bp	321 bp	Not published	STRBase
21.2	344 bp	322 bp	Not published	Allor et al. (2005)
22	346 bp	324 bp	$[AGAA]_{22}$	Barber & Parkin (1996)
22.1	347 bp	325 bp	Not published	STRBase
22.2	348 bp	327 bp	Not published	STRBase
23	350 bp	328 bp	$[AGAA]_{23}$	Barber & Parkin (1996)
23.1	351 bp	329 bp	Not published	SGM Plus
23.2	352 bp	330 bp	Not published	STRBase
24	354 bp	332 bp	$[AGAA]_{24}$	Barber & Parkin (1996)
24.2	356 bp	334 bp	Not published	STRBase
25	358 bp	336 bp	$[AGAA]_{25}$	Barber & Parkin (1996)
26	362 bp	340 bp	$[AGAA]_{26}$	Barber & Parkin (1996)
27	366 bp	344 bp	$[AGAA]_{27}$	Barber & Parkin (1996)
28	370 bp	348 bp	$[AGAA]_{28}$	Morales-Valverde et al. (2009)

(Continued)

D18S51 (*Continued*)

Allele (Repeat #)	Promega PP 16	ABI Identifiler	Repeat Structure $[AGAA]_n$ 3′ flanking region	Reference
28.1	371 bp	349 bp	Not published	Allor et al. (2005)
28.3	373 bp	351 bp	Not published	STRBase
29	374 bp	352 bp	$[AGAA]_{29}$	Morales-Valverde et al. (2009)
29.3	377 bp	355 bp	Not published	STRBase
30	378 bp	356 bp	$[AGAA]_{30}$	Morales-Valverde et al. (2009)
31	382 bp	360 bp	$[AGAA]_{31}$	Morales-Valverde et al. (2009)
32	386 bp	364 bp	$[AGAA]_{32}$	Morales-Valverde et al. (2009)
33	390 bp	368 bp	$[AGAA]_{33}$	Morales-Valverde et al. (2009)
34	394 bp	372 bp	$[AGAA]_{34}$	Morales-Valverde et al. (2009)
35	398 bp	376 bp	$[AGAA]_{35}$	Morales-Valverde et al. (2009)
36	402 bp	380 bp	$[AGAA]_{36}$	Morales-Valverde et al. (2009)
37	406 bp	384 bp	$[AGAA]_{37}$	Morales-Valverde et al. (2009)
38	410 bp	388 bp	$[AGAA]_{38}$	Morales-Valverde et al. (2009)
40	418 bp	396 bp	$[AGAA]_{40}$	Morales-Valverde et al. (2009)

73 observed alleles
**x.2 alleles involve a replacement of the 3′ flanking sequence AAAG with [AGAG]AG (see Dauber et al. 2009)*
***x.1 alleles involve a replacement of the 3′ flanking sequence AAAG with AAAAG (see Dauber et al. 2009)*

D21S11

GenBank Accession: AP000433 (allele 29.1). PCR product sizes of observed alleles

Allele (Repeat #)	**Promega PP16**	**ABI Identifiler**	**Repeat Structure [TCTA]$_n$[TCTG]$_n${[TCTA]$_3$TA[TCTA]$_3$TCA [TCTA]$_2$TCCATA} [TCTA]$_n$ TA TCTA**	**Reference**
12	155 bp	138 bp	Not published	Ayres et al. (2002)
24	203 bp	186 bp	[TCTA]$_4$[TCTG]$_6${43 bp}[TCTA]$_6$	Griffiths et al. (1998)
24.2	205 bp	188 bp	[TCTA]$_5$[TCTG]$_6$ {[TCTA]$_3$ TCA [TCTA]$_2$ TCCA TA}[TCTA]$_9$	Griffiths et al. (1998)
24.3 or **28(D11–13 bp-del)**	206 bp	189 bp	[TCTA]$_5$[TCTG]$_6${43 bp}[TCTA]$_9$ with 13 bp deletion 11 bp downstream	Dauber et al. (2009)
25	207 bp	190 bp	[TCTA]$_4$[TCTG]$_3${43 bp}[TCTA]$_{10}$	Schwartz et al. (1996)
25.2	209 bp	192 bp	[TCTA]$_5$[TCTG]$_6$ {[TCTA]$_3$ TCA [TCTA]$_2$ TCCA TA}[TCTA]$_{10}$	Griffiths et al. (1998)
25.3	210 bp	193 bp	Not published	Allor et al. (2005)
26	211 bp	194 bp	[TCTA]$_4$[TCTG]$_6${43 bp}[TCTA]$_8$	Möller et al. (1994)
26.1	212 bp	195 bp	Not published	SGM Plus
26.2	213 bp	196 bp	Not published	STRBase
27 (a)	215 bp	198 bp	[TCTA]$_4$[TCTG]$_6${43 bp}[TCTA]$_9$	Möller et al. (1994)
27 (b)	215 bp	198 bp	[TCTA]$_6$[TCTG]$_5${43 bp}[TCTA]$_8$	Schwartz et al. (1996)
27 (c)	215 bp	198 bp	[TCTA]$_5$[TCTG]$_5${43 bp}[TCTA]$_9$	Griffiths et al. (1998)
27.1	216 bp	199 bp	Not published	Allor et al. (2005)
27.2	217 bp	200 bp	Not published	STRBase
27.3	218 bp	201 bp	Not published	SGM Plus
28 (a)	219 bp	202 bp	[TCTA]$_4$[TCTG]$_6${43 bp}[TCTA]$_{10}$	Möller et al. (1994)
28 (b)	219 bp	202 bp	[TCTA]$_5$[TCTG]$_6${43 bp}[TCTA]$_9$	Zhou et al. (1997)
28.2 (a)	221 bp	204 bp	[TCTA]$_4$[TCTG]$_6${43 bp}[TCTA]$_{10}$	Griffiths et al. (1998)
28.2 (b)	221 bp	204 bp	[TCTA]$_5$[TCTG]$_6${43 bp}[TCTA]$_8$ TA TCTA	Zhou et al. (1997)
28.3	222 bp	205 bp	Not published	SGM Plus
29 (a)	223 bp	206 bp	[TCTA]$_4$[TCTG]$_6${43 bp}[TCTA]$_{11}$	Griffiths et al. (1998)
29 (b)	223 bp	206 bp	[TCTA]$_6$[TCTG]$_5${43 bp}[TCTA]$_{10}$	Zhou et al. (1997)
29.1	224 bp	207 bp	Not published	Allor et al. (2005)
29.2	225 bp	208 bp	[TCTA]$_5$[TCTG]$_5${43 bp}[TCTA]$_{10}$ TA TCTA	Zhou et al. (1997)
29.3	226 bp	209 bp	Not published	Amorim et al. (2001)

(Continued)

D21S11 (*Continued*)

Allele (Repeat #)	Promega PP16	ABI Identifiler	Repeat Structure [TCTA]$_n$[TCTG]$_n${[TCTA]$_3$TA[TCTA]$_3$TCA [TCTA]$_2$TCCATA} [TCTA]$_n$ TA TCTA	Reference
30 (a)	227 bp	210 bp	[TCTA]$_4$[TCTG]$_6${43 bp}[TCTA]$_{12}$	Schwartz et al. (1996)
30 (b)	227 bp	210 bp	[TCTA]$_5$[TCTG]$_6${43 bp}[TCTA]$_{11}$	Zhou et al. (1997)
30 (c)	227 bp	210 bp	[TCTA]$_6$[TCTG]$_5${43 bp}[TCTA]$_{11}$	Griffiths et al. (1998)
30 (d)	227 bp	210 bp	[TCTA]$_6$[TCTG]$_6${43 bp}[TCTA]$_{10}$	Brinkmann et al. (1996a)
30.1	228 bp	211 bp	Not published	SGM Plus
30.2 (a)	229 bp	212 bp	[TCTA]$_5$[TCTG]$_6${43 bp}[TCTA]$_{10}$ TA TCTA	Griffiths et al. (1998)
30.2 (b)	229 bp	212 bp	[TCTA]$_5$[TCTG]$_5${43 bp}[TCTA]$_{11}$ TA TCTA	Schwartz et al. (1996)
30.3	230 bp	213 bp	[TCTA]$_6$[TCTG]$_5${43 bp}[TCTA]$_5$ TCA [TCTA]$_6$	Tsuji et al. (2006)
31 (a)	231 bp	214 bp	[TCTA]$_5$[TCTG]$_6${43 bp}[TCTA]$_{12}$	Griffiths et al. (1998)
31 (b)	231 bp	214 bp	[TCTA]$_6$[TCTG]$_5${43 bp}[TCTA]$_{12}$	Möller et al. (1994)
31 (c)	231 bp	214 bp	[TCTA]$_6$[TCTG]$_6${43 bp}[TCTA]$_{11}$	Zhou et al. (1997)
31 (d)	231 bp	214 bp	[TCTA]$_7$[TCTG]$_5${43 bp}[TCTA]$_{11}$	Schwartz et al. (1996)
31.1	232 bp	215 bp	Not published	Allor et al. (2005)
31.2	233 bp	216 bp	[TCTA]$_5$[TCTG]$_6${43 bp}[TCTA]$_{11}$ TA TCTA	Griffiths et al. (1998)
31.3	234 bp	217 bp	Not published	SGM Plus
32 (a)	235 bp	218 bp	[TCTA]$_6$[TCTG]$_5${43 bp}[TCTA]$_{13}$	Griffiths et al. (1998)
32 (b)	235 bp	218 bp	[TCTA]$_5$[TCTG]$_6${43 bp}[TCTA]$_{13}$	Zhou et al. (1997)
32.1	236 bp	219 bp	Not published	Steinlechner et al. (2002)
32.2 (a)	237 bp	220 bp	[TCTA]$_5$[TCTG]$_6${43 bp}[TCTA]$_{12}$ TA TCTA	Griffiths et al. (1998)
32.2 (b)	237 bp	220 bp	[TCTA]$_4$[TCTG]$_6${43 bp}[TCTA]$_{13}$ TA TCTA	Brinkmann et al. (1996a)
32.2 (c)	237 bp	220 bp	[TCTA]$_5$[TCTG]$_6${[TCTA]$_2$TA[TCTA]$_3$TCA [TCTA]$_2$TCCATA} [TCTA]$_{13}$ TATCTA	Brinkmann et al. (1996a)
32.3	238 bp	221 bp	Not published	SGM Plus
33	239 bp	222 bp	[TCTA]$_5$[TCTG]$_6${43 bp}[TCTA]$_{14}$	Zhou et al. (1997)
33.1	240 bp	223 bp	Not published	Steinlechner et al. (2002)
33.2 (a)	241 bp	224 bp	[TCTA]$_5$[TCTG]$_6${43 bp}[TCTA]$_{13}$ TA TCTA	Griffiths et al. (1998)
33.2 (b)	241 bp	224 bp	[TCTA]$_6$[TCTG]$_5${43 bp}[TCTA]$_{13}$ TA TCTA	Brinkmann et al. (1996a)
33.2 (c)	241 bp	224 bp	[TCTA]$_6$[TCTG]$_6${43 bp}[TCTA]$_{12}$ TA TCTA	Brinkmann et al. (1996a)

(Continued)

D21S11 (*Continued*)

Allele (Repeat #)	Promega PP16	ABI Identifiler	Repeat Structure $[TCTA]_n[TCTG]_n\{[TCTA]_3TA[TCTA]_3TCA[TCTA]_2TCCATA\}[TCTA]_n$ TA TCTA	Reference
33.3	242 bp	225 bp	$[TCTA]_5[TCTG]_6\{43\,bp\}[TCTA]_8TCA[TCTA]_3TCA[TCTA]_2TA$ TCTA	Brinkmann et al. (1996a)
34 (a)	243 bp	226 bp	$[TCTA]_5[TCTG]_6\{43\,bp\}[TCTA]_{15}$	Zhou et al. (1997)
34 (b)	243 bp	226 bp	$[TCTA]_{10}[TCTG]_5\{43\,bp\}[TCTA]_{11}$	Brinkmann et al. (1996a)
34.1	244 bp	227 bp	Not published	Ayres et al. (2002)
34.2	245 bp	228 bp	$[TCTA]_5[TCTG]_6\{43\,bp\}[TCTA]_{14}TATCTA$	Griffiths et al. (1998)
34.3	246 bp	229 bp	$[TCTA]_5[TCTG]_6\{43\,bp\}[TCTA]_{10}TCA[TCTA]_4TATCTA$	Brinkmann et al. (1996a)
35 (a)	247 bp	230 bp	$[TCTA]_{10}[TCTG]_5\{43\,bp\}[TCTA]_{12}$	Griffiths et al. (1998)
35 (b)	247 bp	230 bp	$[TCTA]_{11}[TCTG]_5\{43\,bp\}[TCTA]_{11}$	Brinkmann et al. (1996a)
35.1	248 bp	231 bp	Not published	Steinlechner et al. (2002)
35.2	249 bp	232 bp	$[TCTA]_5[TCTG]_6\{43\,bp\}[TCTA]_{15}$ TA TCTA	Zhou et al. (1997)
35.3	250 bp	233 bp	Not published	SGM Plus
36 (a)	251 bp	234 bp	$[TCTA]_{11}[TCTG]_5\{43\,bp\}[TCTA]_{12}$	Griffiths et al. (1998)
36 (b)	251 bp	234 bp	$[TCTA]_{10}[TCTG]_5\{43\,bp\}[TCTA]_{13}$	Brinkmann et al. (1996a)
36 (c)	251 bp	234 bp	$[TCTA]_{10}[TCTG]_6\{43\,bp\}[TCTA]_{12}$	Brinkmann et al. (1996a)
36.1	252 bp	235 bp	Not published	Allor et al. (2005)
36.2	253 bp	236 bp	$[TCTA]_5[TCTG]_6\{43\,bp\}[TCTA]_{16}$ TA TCTA	Zhou et al. (1997)
36.3	254 bp	237 bp	Not published	SGM Plus
37 (a)	255 bp	238 bp	$[TCTA]_{11}[TCTG]_5\{43\,bp\}[TCTA]_{13}$	Griffiths et al. (1998)
37 (b)	255 bp	238 bp	$[TCTA]_9[TCTG]_{11}\{43\,bp\}[TCTA]_{12}$	Brinkmann et al. (1996a)
37.2 (a)	257 bp	240 bp	$[TCTA]_7[TCTG]_{14}\{[TCTA]_3$ TCA $[TCTA]_2$ TCCA TA$\}[TCTA]_{12}$	Walsh et al. (2003)
37.2 (b)	257 bp	240 bp	$[TCTA]_9[TCTG]_{12}\{[TCTA]_3$ TCA $[TCTA]_2$ TCCA TA$\}[TCTA]_{12}$	Walsh et al. (2003)
37.2 (c)	257 bp	240 bp	$[TCTA]_9[TCTG]_{13}\{[TCTA]_3$ TCA $[TCTA]_2$ TCCA TA$\}[TCTA]_{11}$	Walsh et al. (2003)
37.2 (d)	257 bp	240 bp	$[TCTA]_{10}[TCTG]_{11}\{[TCTA]_3$ TCA $[TCTA]_2$ TCCA TA$\}[TCTA]_{12}$	Walsh et al. (2003)
37.2 (e)	257 bp	240 bp	$[TCTA]_{11}[TCTG]_{11}\{[TCTA]_3$ TCA $[TCTA]_2$ TCCA TA$\}[TCTA]_{11}$	Walsh et al. (2003)
38 (a)	259 bp	242 bp	$[TCTA]_{13}[TCTG]_5\{43\,bp\}[TCTA]_{12}$	Griffiths et al. (1998)
38 (b)	259 bp	242 bp	$[TCTA]_9[TCTG]_{11}\{43\,bp\}[TCTA]_{12}$	Brinkmann et al. (1996a)

(Continued)

D21S11 (*Continued*)

Allele (Repeat #)	Promega PP16	ABI Identifiler	Repeat Structure [TCTA]$_n$[TCTG]$_n${[TCTA]$_3$TA[TCTA]$_3$TCA [TCTA]$_2$TCCATA} [TCTA]$_n$ TA TCTA	Reference
38 (c)	259 bp	242 bp	[TCTA]$_{10}$[TCTG]$_{11}${43 bp}[TCTA]$_{13}$	Brinkmann et al. (1996a)
38 (d)	259 bp	242 bp	[TCTA]$_{11}$[TCTG]$_{11}${43 bp}[TCTA]$_{11}$	Brinkmann et al. (1996a)
38.2 (a)	261 bp	244 bp	[TCTA]$_9$[TCTG]$_{12}${[TCTA]$_3$ TCA [TCTA]$_2$ TCCA TA}[TCTA]$_{13}$	Walsh et al. (2003)
38.2 (b)	261 bp	244 bp	[TCTA]$_9$[TCTG]$_{13}${[TCTA]$_3$ TCA [TCTA]$_2$ TCCA TA}[TCTA]$_{12}$	Walsh et al. (2003)
38.2 (c)	261 bp	244 bp	[TCTA]$_{10}$[TCTG]$_{11}${[TCTA]$_3$ TCA [TCTA]$_2$ TCCA TA}[TCTA]$_{13}$	Walsh et al. (2003)
39	263 bp	246 bp	Not published	STRBase
39.2 (a)	265 bp	248 bp	[TCTA]$_{10}$[TCTG]$_{13}${[TCTA]$_3$ TCA [TCTA]$_2$ TCCA TA}[TCTA]$_{12}$	Bagdonavicius et al. (2002); Walsh et al. (2003)
39.2 (b)	265 bp	248 bp	[TCTA]$_{11}$[TCTG]$_{12}${[TCTA]$_3$ TCA [TCTA]$_2$ TCCA TA}[TCTA]$_{12}$	Bagdonavicius et al. (2002); Walsh et al. (2003)
40.2	269 bp	252 bp	Not published	Ayres et al. (2002)
41.2	273 bp	256 bp	[TCTA]$_{10}$[TCTG]$_{15}${[TCTA]$_3$ TCA [TCTA]$_2$ TCCA TA}[TCTA]$_{12}$	Bagdonavicius et al. (2002); Walsh et al. (2003)
43.2	281 bp	264 bp	[TCTA]$_5$[TCTG]$_6${[TCTA]$_3$ TA [TCTA]$_3$ TCA [TCTA]$_2$ TCCA TA}[TCTA]$_8$ TCA [TCTA]$_2$ TCCA TA[TCTA]$_{13}$ TA TCTA	Grubwieser et al. (2005)

90 observed alleles

D2S1338

GenBank Accession: AC010136 (allele 23). PCR product sizes of observed alleles

Allele (Repeat #)	Promega ESX 17	Promega ESI 17	ABI Identifiler*	Repeat Structure $[TGCC]_m[TTCC]_n$	Reference
10	197 bp	223 bp	287 bp	$[TGCC]_4[TTCC]_6$	Heinrich et al. (2005)
11	201 bp	227 bp	291 bp	Not published	STRBase
12 (a)	205 bp	231 bp	295 bp	$[TGCC]_4[TTCC]_8$	Kline et al. (2010)
12 (b)	205 bp	231 bp	295 bp	$[TGCC]_6[TTCC]_6$	Kline et al. (2010)
12.3	208 bp	234 bp	298 bp	Not published	STRBase
13	209 bp	235 bp	299 bp	$[TGCC]_6[TTCC]_7$	Kline et al. (2010)
14	213 bp	239 bp	303 bp	Not published	STRBase
15	217 bp	243 bp	307 bp	Not published	SGM Plus
16	221 bp	247 bp	311 bp	Not published	SGM Plus
17	225 bp	251 bp	315 bp	$[TGCC]_6[TTCC]_{11}$	Heinrich et al. (2005)
18	229 bp	255 bp	319 bp	$[TGCC]_6[TTCC]_{12}$	Heinrich et al. (2005)
18.3	232 bp	258 bp	322 bp	Not published	STRBase
19 (a)	233 bp	259 bp	323 bp	$[TGCC]_8[TTCC]_{11}$	Heinrich et al. (2005)
19 (b)	233 bp	259 bp	323 bp	$[TGCC]_7[TTCC]_{12}$	Heinrich et al. (2005)
19.1	234 bp	260 bp	324 bp	Not published	STRBase
19.3	236 bp	262 bp	326 bp	Not published	STRBase
20 (a)	237 bp	263 bp	327 bp	$[TGCC]_7TCCC[TTCC]_{12}$	Heinrich et al. (2005)
20 (b)	237 bp	263 bp	327 bp	$[TGCC]_7[TTCC]_{10}GTCC[TTCC]_2$	Heinrich et al. (2005)
20.1	238 bp	264 bp	328 bp	Not published	STRBase
21	241 bp	267 bp	331 bp	$[TGCC]_7[TTCC]_{11}GTCC[TTCC]_2$	Heinrich et al. (2005)
21.2	243 bp	269 bp	333 bp	Not published	Montelius et al. (2008)
22	245 bp	271 bp	335 bp	$[TGCC]_7[TTCC]_{12}GTCC[TTCC]_2$	Heinrich et al. (2005)
22.2	247 bp	273 bp	337 bp	Not published	STRBase
22.3	248 bp	274 bp	338 bp	Not published	STRBase
23 (a)	249 bp	275 bp	339 bp	$[TGCC]_7[TTCC]_{13}GTCC[TTCC]_2$	Heinrich et al. (2005)
23 (b)	249 bp	275 bp	339 bp	$[TGCC]_7[TTCC]_{13}GTCC\ [TTCC]_2$	GenBank allele
23.2	251 bp	277 bp	341 bp	Not published	STRBase
23.3	252 bp	278 bp	342 bp	$[TGCC]_7[TTCC]_4TTC[TTCC]_9$ $GTCC[TTCC]_2$	Kline et al. (2010)
24	253 bp	279 bp	343 bp	$[TGCC]_7[TTCC]_{14}GTCC[TTCC]_2$	Heinrich et al. (2005)
24.2	255 bp	281 bp	345 bp	Not published	STRBase

(Continued)

D2S1338 (*Continued*)

Allele (Repeat #)	Promega ESX 17	Promega ESI 17	ABI Identifiler*	Repeat Structure $[TGCC]_m[TTCC]_n$	Reference
25	257 bp	283 bp	347 bp	$[TGCC]_7[TTCC]_{15}GTCC[TTCC]_2$	Heinrich et al. (2005)
25.2	259 bp	285 bp	349 bp	Not published	STRBase
25.3	260 bp	286 bp	350 bp	Not published	STRBase
26 (a)	261 bp	287 bp	351 bp	$[TGCC]_7[TTCC]_{16}GTCC[TTCC]_2$	Heinrich et al. (2005)
26 (b)	261 bp	287 bp	351 bp	$[TGCC]_7[TTCC]_{19}$	Kline et al. (2010)
27	265 bp	291 bp	355 bp	$[TGCC]_7[TTCC]_{17}GTCC[TTCC]_2$	Heinrich et al. (2005)
28	269 bp	295 bp	359 bp	Not published	SGM Plus
29.3	276 bp	302 bp	366 bp	Not published	STRBase
30.3	280 bp	306 bp	370 bp	Not published	STRBase
31	281 bp	307 bp	371 bp	$[TGCC]_7[TTCC]_6TTAC[TTCC]_{14}GTCC[TTCC]_2$	Kline et al. (2010)

40 observed alleles
**Mobility modified PCR products so observed size will not equal expected DNA sequence size*

D19S433

GenBank Accession: AC008507 (allele 14). PCR product sizes of observed alleles

Allele (Repeat #)	Promega ESX 17	Promega ESI 17	ABI Identifiler	Repeat Structure (AAGG)AAAG(AAGG) TAGG[AAGG]$_n$	Reference
5.2	193 bp	163 bp	92 bp	Not published	STRBase
6.2	197 bp	167 bp	96 bp	(AAGG)AA–(AAGG) TAGG[AAGG]$_5$	Heinrich et al. (2005)
7	199 bp	169 bp	98 bp	Not published	STRBase
8	203 bp	173 bp	102 bp	Not published	STRBase
8.1	204 bp	174 bp	103 bp	Not published	STRBase
9	207 bp	177 bp	106 bp	(AAGG)AAAG(AAGG) TAGG(AAGA)[AAGG]$_6$	Heinrich et al. (2005)
9.1	208 bp	178 bp	107 bp	Not published	STRBase
10	211 bp	181 bp	110 bp	(AAGG)AAAG(AAGG) TAGG[AAGG]$_8$	Heinrich et al. (2005)
10.1	212 bp	182 bp	111 bp	Not published	STRBase
10.2	213 bp	183 bp	112 bp	Not published	STRBase
11	215 bp	185 bp	114 bp	(AAGG)AAAG(AAGG) TAGG[AAGG]$_9$	Heinrich et al. (2005)
11.1	216 bp	186 bp	115 bp	(AAGG)AAAG(AAGG) TAGG(AAGG)A[AAGG]$_8$	Heinrich et al. (2005)
11.2	217 bp	187 bp	116 bp	Not published	STRBase
12	219 bp	189 bp	118 bp	(AAGG)AAAG(AAGG) TAGG[AAGG]$_{10}$	Heinrich et al. (2005)
12.1	220 bp	190 bp	119 bp	(AAGG)AAAG(AAGG) TAGG[AAGG]$_5$A[AAGG]$_5$	Heinrich et al. (2005)
12.2	221 bp	191 bp	120 bp	Not published	STRBase
13	223 bp	193 bp	122 bp	(AAGG)AAAG(AAGG) TAGG[AAGG]$_{11}$	Heinrich et al. (2005)
13.1	224 bp	194 bp	123 bp	Not published	STRBase
13.2	225 bp	195 bp	124 bp	(AAGG)AA–(AAGG) TAGG[AAGG]$_{12}$	Heinrich et al. (2005)
13.3	226 bp	196 bp	125 bp	Not published	STRBase
14	227 bp	197 bp	126 bp	(AAGG)AAAG(AAGG) TAGG[AAGG]$_{12}$	Heinrich et al. (2005)
14.1	228 bp	198 bp	127 bp	Not published	STRBase
14.2	229 bp	199 bp	128 bp	Not published	STRBase
14.3	230 bp	200 bp	129 bp	Not published	STRBase

(Continued)

D19S433 (*Continued*)

Allele (Repeat #)	Promega ESX 17	Promega ESI 17	ABI Identifiler	Repeat Structure (AAGG)AAAG(AAGG) TAGG[AAGG]$_n$	Reference
15	231 bp	201 bp	130 bp	(AAGG)AAAG(AAGG) TAGG[AAGG]$_{13}$	Heinrich et al. (2005)
15.2	233 bp	203 bp	132 bp	(AAGG)AA–(AAGG) TAGG[AAGG]$_{14}$	Heinrich et al. (2005)
16	235 bp	205 bp	134 bp	(AAGG)AAAG(AAGG) TAGG[AAGG]$_{14}$	Heinrich et al. (2005)
16.2	237 bp	207 bp	136 bp	(AAGG)AA–(AAGG) TAGG[AAGG]$_{15}$	Heinrich et al. (2005)
17	239 bp	209 bp	138 bp	(AAGG)AAAG(AAGG) TAGG[AAGG]$_{15}$	Heinrich et al. (2005)
17.2	241 bp	211 bp	140 bp	(AAGG)AA–(AAGG) TAGG[AAGG]$_{16}$	Heinrich et al. (2005)
18	243 bp	213 bp	142 bp	(AAGG)AAAG(AAGG) TAGG[AAGG]$_{16}$	Heinrich et al. (2005)
18.1	244 bp	214 bp	143 bp	Not published	STRBase
18.2	245 bp	215 bp	144 bp	Not published	STRBase
19	247 bp	217 bp	146 bp	Not published	STRBase
19.2	249 bp	219 bp	148 bp	AA–(AAGG)TAGG[AAGG]$_{18}$	M. Kline (unpublished)
20	251 bp	221 bp	150 bp	Not published	STRBase

36 observed alleles

Penta D

GenBank Accession: AP001752 (allele 13). PCR product sizes of observed alleles

Allele (Repeat #)	Promega PP16	Repeat Structure $[AAAGA]_n$	Reference
1.1	370 bp	Not published	STRBase
1.2	371 bp	Not published	STRBase
2.2	376 bp	$[AAAGA]_5$ (U13-13 bp-del)	Bacher et al. (1998)
3	379 bp	Not published	STRBase
3.2	381 bp	$[AAAGA]_5$ (U8-8 bp-del)	Bacher et al. (1998)
4	384 bp	Not published	STRBase
4.1	385 bp	Not published	STRBase
5	389 bp	$[AAAGA]_5$	Krenke et al. (2002)
5.4	393 bp	Not published	STRBase
6	394 bp	$[AAAGA]_6$	Kline et al. (2010)
6.4	398 bp	Not published	STRBase
7	399 bp	$[AAAGA]_7$	Krenke et al. (2002)
7.1	400 bp	Not published	STRBase
7.4	403 bp	$[AAAGA]_8$ (U1A-del)	Kline et al. (2010)
8	404 bp	$[AAAGA]_8$	Krenke et al. (2002)
8.1	405 bp	Not published	STRBase
8.2	406 bp	$[AAAGA]_{11}$ (U13-13 bp-del)	Kline et al. (2010)
8.4	408 bp	Not published	STRBase
9	409 bp	$[AAAGA]_9$	Kline et al. (2010)
9.1	410 bp	Not published	STRBase
9.2	411 bp	$[AAAGA]_{10}$ (D128TAA-del)	Miozzo et al. (2007)
9.4	413 bp	Not published	STRBase
10	414 bp	$[AAAGA]_{10}$	Kline et al. (2010)
10.1	415 bp	Not published	STRBase
10.2	416 bp	Not published	STRBase
10.3	417 bp	Not published	STRBase
11	419 bp	$[AAAGA]_{11}$	Kline et al. (2010)
11.1	420 bp	Not published	STRBase
11.2	421 bp	Not published	Huel et al. (2007)

(Continued)

Penta D (*Continued*)

Allele (Repeat #)	Promega PP16	Repeat Structure $[AAAGA]_n$	Reference
11.3	422 bp	Not published	STRBase
11.4	423 bp	Not published	STRBase
12	424 bp	$[AAAGA]_{12}$	Krenke et al. (2002)
12.1	425 bp	Not published	Huel et al. (2007)
12.2	426 bp	Not published	Huel et al. (2007)
12.3	427 bp	Not published	STRBase
12.4	428 bp	Not published	STRBase
13	429 bp	$[AAAGA]_{13}$	Krenke et al. (2002)
13.2	431 bp	Not published	STRBase
13.3	432 bp	Not published	Bacher et al. (1998)
13.4	433 bp	Not published	STRBase
14	434 bp	$[AAAGA]_{14}$	Krenke et al. (2002)
14.1	435 bp	Not published	Huel et al. (2007)
14.4	438 bp	Not published	STRBase
15	439 bp	$[AAAGA]_{15}$	Krenke et al. (2002)
15.1	440 bp	Not published	STRBase
15.4	443 bp	Not published	STRBase
16	444 bp	$[AAAGA]_{16}$	Krenke et al. (2002)
17	449 bp	$[AAAGA]_{17}$	Krenke et al. (2002)
18	454 bp	$[AAAGA]_{18}$	Kline et al. (2010)
19	459 bp	$[AAAGA]_{19}$	Kline et al. (2010)

50 observed alleles

Penta E

GenBank Accession: AC027004 (allele 5). PCR product sizes of observed alleles

Allele (Repeat #)	Promega PP16	Repeat Structure $[AAAGA]_n$	Reference
5	379 bp	$[AAAGA]_5$	Krenke et al. (2002)
5.1	380 bp	Not published	STRBase
6	384 bp	$[AAAGA]_6$	Krenke et al. (2002)
7	389 bp	$[AAAGA]_7$	Krenke et al. (2002)
8	394 bp	$[AAAGA]_8$	Krenke et al. (2002)
9	399 bp	$[AAAGA]_9$	Krenke et al. (2002)
9.1	400 bp	Not published	STRBase
9.2	401 bp	Not published	STRBase
9.4	403 bp	Not published	STRBase
10	404 bp	$[AAAGA]_{10}$	Krenke et al. (2002)
10.1	405 bp	Not published	STRBase
10.2	406 bp	Not published	STRBase
11	409 bp	$[AAAGA]_{11}$	Krenke et al. (2002)
11.4	413 bp	Not published	STRBase
12	414 bp	$[AAAGA]_{12}$	Krenke et al. (2002)
12.1	415 bp	Not published	STRBase
12.2	416 bp	Not published	STRBase
12.3	417 bp	Not published	STRBase
13	419 bp	$[AAAGA]_{13}$	Krenke et al. (2002)
13.2	421 bp	Not published	STRBase
13.4	423 bp	Not published	STRBase
14	424 bp	$[AAAGA]_{14}$	Krenke et al. (2002)
14.2	426 bp	Not published	STRBase
14.4	428 bp	Not published	STRBase
15	429 bp	$[AAAGA]_{15}$	Krenke et al. (2002)
15.2	431 bp	Not published	STRBase
15.4	433 bp	Not published	STRBase
16	434 bp	$[AAAGA]_{16}$	Krenke et al. (2002)
16.2	436 bp	Not published	STRBase
16.4	438 bp	Not published	Huel et al. (2007)

(Continued)

Penta E (*Continued*)

Allele (Repeat #)	Promega PP16	Repeat Structure $[AAAGA]_n$	Reference
17	439 bp	$[AAAGA]_{17}$	Krenke et al. (2002)
17.2	441 bp	Not published	STRBase
17.4	443 bp	Not published	Huel et al. (2007)
18	444 bp	$[AAAGA]_{18}$	Krenke et al. (2002)
18.2	446 bp	Not published	Huel et al. (2007)
18.4	448 bp	$[AAAGA]_6AAAA[AAAGA]_{12}$	Mizuno et al. (2003)
19	449 bp	$[AAAGA]_{19}$	Krenke et al. (2002)
19.2	451 bp	Not published	STRBase
19.4	453 bp	$[AAAGA]_6AAAA[AAAGA]_{13}$	Mizuno et al. (2003)
20	454 bp	$[AAAGA]_{20}$	Krenke et al. (2002)
20.2	456 bp	Not published	STRBase
21	459 bp	$[AAAGA]_{21}$	Krenke et al. (2002)
22	464 bp	$[AAAGA]_{22}$	Krenke et al. (2002)
23	469 bp	$[AAAGA]_{23}$	Krenke et al. (2002)
23.4	473 bp	Not published	STRBase
24	474 bp	$[AAAGA]_{24}$	Krenke et al. (2002)
26	484 bp	Not published	STRBase
26.2	486 bp	Not published	STRBase
27	489 bp	$[AAAGA]_{27}$	Kline et al. (2010)
29	499 bp	$[AAAGA]_{29}$	Kline et al. (2010)
30	504 bp	$[AAAGA]_{30}$	Kline et al. (2010)
31	509 bp	$[AAAGA]_{31}$	Kline et al. (2010)
32	514 bp	$[AAAGA]_{32}$	Kline et al. (2010)

53 observed alleles

D1S1656

GenBank Accession: G07820 (allele 15.3). PCR product sizes of observed alleles

Allele (Repeat #)	Promega ESX 17	Promega ESI 17	ABI NGM	Repeat Structure $[TAGA]_4[TGA]_{0–1}$ $[TAGA]_nTAGG[TG]_5$	Reference
8	133 bp	222 bp	171 bp	$[TAGA]_8[TG]_5$	Phillips et al. (2011)
9	137 bp	226 bp	175 bp	$[TAGA]_9[TG]_5$	Phillips et al. (2011)
10 (a)	141 bp	230 bp	179 bp	$[TAGA]_{10}[TG]_5$	Lareu et al. (1998)
10 (b)	141 bp	230 bp	179 bp	$[TAGA]_{10}TAGG[TG]_5$	Phillips et al. (2011)
11	145 bp	234 bp	183 bp	$[TAGA]_{11}[TG]_5$	Lareu et al. (1998)
12 (a)	149 bp	238 bp	187 bp	$[TAGA]_{12}[TG]_5$	Lareu et al. (1998)
12 (b)	149 bp	238 bp	187 bp	$[TAGA]_{11}TAGG[TG]_5$	Lareu et al. (1998)
13 (a)	153 bp	242 bp	191 bp	$[TAGA]_{12}TAGG[TG]_5$	Lareu et al. (1998)
13 (b)	153 bp	242 bp	191 bp	$[TAGA]_{13}[TG]_5$	Phillips et al. (2011)
13.3	156 bp	245 bp	194 bp	$[TAGA]_1TGA[TAGA]_{11}TAGG[TG]_5$	Phillips et al. (2011)
14 (a)	157 bp	246 bp	195 bp	$[TAGA]_{13}TAGG[TG]_5$	Lareu et al. (1998)
14 (b)	157 bp	246 bp	195 bp	$[TAGA]_{14}[TG]_5$	Phillips et al. (2011)
14.3	160 bp	249 bp	198 bp	$[TAGA]_4TGA[TAGA]_9TAGG[TG]_5$	Phillips et al. (2011)
15	161 bp	250 bp	199 bp	$[TAGA]_{14}TAGG[TG]_5$	Lareu et al. (1998)
15.3	164 bp	253 bp	202 bp	$[TAGA]_4TGA[TAGA]_{10}TAGG[TG]_5$	Lareu et al. (1998)
16	165 bp	254 bp	203 bp	$[TAGA]_{15}TAGG[TG]_5$	Lareu et al. (1998)
16.3	168 bp	257 bp	206 bp	$[TAGA]_4TGA[TAGA]_{11}TAGG[TG]_5$	Lareu et al. (1998)
17	169 bp	258 bp	207 bp	$[TAGA]_{16}TAGG[TG]_5$	Lareu et al. (1998)
17.1	170 bp	259 bp	208 bp	Not published	Schröer et al. (2000)
17.3	172 bp	261 bp	210 bp	$[TAGA]_4TGA[TAGA]_{12}TAGG[TG]_5$	Lareu et al. (1998)
18	173 bp	262 bp	211 bp	$[TAGA]_{17}TAGG[TG]_5$	Phillips et al. (2011)
18.3	176 bp	265 bp	214 bp	$[TAGA]_4TGA[TAGA]_{13}TAGG[TG]_5$	Lareu et al. (1998)
19	177 bp	266 bp	215 bp	Not published	Asamura et al. (2008)
19.3	180 bp	269 bp	218 bp	$[TAGA]_4TGA[TAGA]_{14}TAGG[TG]_5$	Lareu et al. (1998)
20.3	184 bp	273 bp	222 bp	Not published	Gamero et al. (2000)

25 observed alleles

D2S441

GenBank Accession: AC079112 (allele 12). PCR product sizes of observed alleles

Allele (Repeat #)	Promega ESX 17	Promega ESI 17	ABI NGM	Repeat Structure $[TCTA]_4(TCA)_{0–1}[TCTA]_n[TTTA]_{0–1}[TCTA]_2$	Reference
8	88 bp	347 bp	76 bp	$[TCTA]_8$	Coble & Butler (2005)
9	92 bp	351 bp	80 bp	$[TCTA]_9$	Coble & Butler (2005)
9.1	93 bp	352 bp	81 bp	$A[TCTA]_9$	Hill et al. (2010)
10 (a)	96 bp	355 bp	84 bp	$[TCTA]_{10}$	Coble & Butler (2005)
10 (b)	96 bp	355 bp	84 bp	$[TCTA]_5TCAA[TCTA]_4$	Phillips et al. (2011)
10 (c)	96 bp	355 bp	84 bp	$[TCTA]_5TCAA[TCTA]_2$ TCTGTCTA	Phillips et al. (2011)
10 (d)	96 bp	355 bp	84 bp	$[TCTA]_8TCTGTCTA$	Phillips et al. (2011)
11 (a)	100 bp	359 bp	88 bp	$[TCTA]_{11}$	Coble & Butler (2005)
11 (b)	100 bp	359 bp	88 bp	$[TCTA]_2TCAA[TCTA]_8$	Phillips et al. (2011)
11.3	103 bp	362 bp	91 bp	$[TCTA]_4TCA[TCTA]_7$	Phillips et al. (2011)
12 (a)	104 bp	363 bp	92 bp	$[TCTA]_{12}$	Coble & Butler (2005)
12 (b)	104 bp	363 bp	92 bp	$[TCTA]_5TCAA[TCTA]_6$	Phillips et al. (2011)
12.3	107 bp	366 bp	95 bp	$[TCTA]_4TCATCCA[TCTA]_7$	Phillips et al. (2011)
13	108 bp	367 bp	96 bp	Not published	Coble & Butler (2005)
13.3	111 bp	370 bp	99 bp	Not published	Coble & Butler (2005)
14 (a)	112 bp	371 bp	100 bp	$[TCTA]_{11}TTTA[TCTA]_2$	Phillips et al. (2011)
14 (b)	112 bp	371 bp	100 bp	$[TCTA]_5TCAA[TCTA]_5TTTA[TCTA]_2$	Phillips et al. (2011)
14.3	115 bp	374 bp	103 bp	Not published	Coble & Butler (2005)
15	116 bp	375 bp	104 bp	Not published	Coble & Butler (2005)
16 (a)	120 bp	379 bp	108 bp	$[TCTA]_{13}TTTA[TCTA]_2$	Phillips et al. (2011)
16 (b)	120 bp	379 bp	108 bp	$[TCTA]_5TCAA[TCTA]_7TTTA[TCTA]_2$	Phillips et al. (2011)
17	124 bp	383 bp	112 bp	$[TCTA]_{14}TTTA[TCTA]_2$	Phillips et al. (2011)

22 observed alleles

D10S1248

GenBank Accession: AL391869 (allele 13). PCR product sizes of observed alleles

Allele (Repeat #)	Promega ESX 17	Promega ESI 17	ABI NGM	Repeat Structure $[GGAA]_n$	Reference
7	79 bp	282 bp	73 bp	$[GGAA]_7$	Phillips et al. (2011)
8	83 bp	286 bp	77 bp	$[GGAA]_8$	Coble & Butler (2005)
9	87 bp	290 bp	81 bp	$[GGAA]_9$	Coble & Butler (2005)
10	91 bp	294 bp	85 bp	$[GGAA]_{10}$	Coble & Butler (2005)
11	95 bp	298 bp	89 bp	$[GGAA]_{11}$	Coble & Butler (2005)
12	99 bp	302 bp	93 bp	$[GGAA]_{12}$	Coble & Butler (2005)
13	103 bp	306 bp	97 bp	$[GGAA]_{13}$	Coble & Butler (2005)
14	107 bp	310 bp	101 bp	$[GGAA]_{14}$	Coble & Butler (2005)
15	111 bp	314 bp	105 bp	$[GGAA]_{15}$	Coble & Butler (2005)
16	115 bp	318 bp	109 bp	$[GGAA]_{16}$	Coble & Butler (2005)
17	119 bp	322 bp	113 bp	$[GGAA]_{17}$	Coble & Butler (2005)
18	123 bp	326 bp	117 bp	$[GGAA]_{18}$	Coble & Butler (2005)
19	127 bp	330 bp	121 bp	$[GGAA]_{19}$	Coble & Butler (2005)

13 observed alleles

D12S391

GenBank Accession: G08921 (allele 20). PCR product sizes of observed alleles

Allele (Repeat #)	Promega ESX 17	Promega ESI 17	ABI NGM	Repeat Structure $[AGAT]_n\{GAT\}_{0–1}(AGAC)_m[AGAT]_{0–1}$	Reference
13	126 bp	287 bp	226 bp	Not published	Poetsch et al. (2011)
14	130 bp	291 bp	230 bp	$[AGAT]_8(AGAC)_5AGAT$	Phillips et al. (2011)
15	134 bp	295 bp	234 bp	$[AGAT]_8(AGAC)_6AGAT$	Lareu et al. (1996)
16	138 bp	299 bp	238 bp	$[AGAT]_9(AGAC)_6AGAT$	Lareu et al. (1996)
17 (a)	142 bp	303 bp	242 bp	$[AGAT]_{10}(AGAC)_6AGAT$	Lareu et al. (1996)
17 (b)	142 bp	303 bp	242 bp	$[AGAT]_{11}(AGAC)_5AGAT$	Phillips et al. (2011)
17.1	143 bp	304 bp	243 bp	Not published	Hill et al. (2010)
17.3	145 bp	306 bp	245 bp	$[AGAT]GAT[AGAT]_8(AGAC)_7AGAT$	Phillips et al. (1998)
18	146 bp	307 bp	246 bp	$[AGAT]_{11}(AGAC)_6AGAT$	Lareu et al. (1996)
18.1	147 bp	308 bp	247 bp	Not published	Hill et al. (2010)
18.3	149 bp	310 bp	249 bp	$[AGAT]GAT[AGAT]_9(AGAC)_7AGAT$	Phillips et al. (1998)
19	150 bp	311 bp	250 bp	$[AGAT]_{12}(AGAC)_6AGAT$	Lareu et al. (1996)
19.1	151 bp	312 bp	251 bp	Not published	Hill et al. (2010)
19.2	152 bp	313 bp	252 bp	Not published	Berti et al. (2010)
19.3 (a)	153 bp	314 bp	253 bp	$[AGAT]GAT[AGAT]_{10}(AGAC)_7AGAT$	Phillips et al. (1998)
19.3 (b)	153 bp	314 bp	253 bp	$[AGAT]_5GAT[AGAT]_7(AGAC)_7$	Phillips et al. (1998)
19.3 (c)	153 bp	314 bp	253 bp	$[AGAT]_5GAT[AGAT]_7(AGAC)_6AGAT$	Phillips et al. (2011)
20 (a)	154 bp	315 bp	254 bp	$[AGAT]_{13}(AGAC)_6AGAT$	Lareu et al. (1996)
20 (b)	154 bp	315 bp	254 bp	$[AGAT]_{11}(AGAC)_9$	Lareu et al. (1996)
20 (c)	154 bp	315 bp	254 bp	$[AGAT]_{10}(AGAC)_9AGAT$	M. Kline (unpublished)
20 (d)	154 bp	315 bp	254 bp	$[AGAT]_{12}(AGAC)_7AGAT$	M. Kline (unpublished)
20.1	155 bp	316 bp	255 bp	Not published	Hill et al. (2010)
20.3	157 bp	318 bp	257 bp	Not published	Phillips et al. (1998)
21 (a)	158 bp	319 bp	258 bp	$[AGAT]_{14}(AGAC)_6AGAT$	Lareu et al. (1996)
21 (b)	158 bp	319 bp	258 bp	$[AGAT]_{12}(AGAC)_9$	Lareu et al. (1996)
21 (c)	158 bp	319 bp	258 bp	$[AGAT]_{12}(AGAC)_8AGAT$	Phillips et al. (2011)
21 (d)	158 bp	319 bp	258 bp	$[AGAT]_{13}(AGAC)_7AGAT$	Phillips et al. (2011)
21 (e)	158 bp	319 bp	258 bp	$[AGAT]_{11}(AGAC)_{10}$	M. Kline (unpublished)

(Continued)

D12S391 (*Continued*)

Allele (Repeat #)	Promega ESX 17	Promega ESI 17	ABI NGM	Repeat Structure $[AGAT]_n\{GAT\}_{0–1}(AGAC)_m[AGAT]_{0–1}$	Reference
22 (a)	162 bp	323 bp	262 bp	$[AGAT]_{15}(AGAC)_6AGAT$	Lareu et al. (1996)
22 (b)	162 bp	323 bp	262 bp	$[AGAT]_{12}(AGAC)_{10}$	Lareu et al. (1996)
22 (c)	162 bp	323 bp	262 bp	$[AGAT]_{11}(AGAC)_{11}$	Phillips et al. (2011)
22 (d)	162 bp	323 bp	262 bp	$[AGAT]_{13}(AGAC)_9$	M. Kline (unpublished)
22.2	164 bp	325 bp	264 bp	Not published	Hill et al. (2010)
22.3	165 bp	326 bp	265 bp	$[AGAT]_3GAT[AGAT]_{10}(AGAC)_9$	Phillips et al. (2011)
23 (a)	166 bp	327 bp	266 bp	$[AGAT]_{14}(AGAC)_8AGAT$	Lareu et al. (1996)
23 (b)	166 bp	327 bp	266 bp	$[AGAT]_{14}(AGAC)_9$	Lareu et al. (1996)
24 (a)	170 bp	331 bp	270 bp	$[AGAT]_{15}(AGAC)_8AGAT$	Lareu et al. (1996)
24 (b)	170 bp	331 bp	270 bp	$[AGAT]_{14}(AGAC)_9AGAT$	Lareu et al. (1996)
24 (c)	170 bp	331 bp	270 bp	$[AGAT]_{15}(AGAC)_9$	Lareu et al. (1996)
24 (d)	170 bp	331 bp	270 bp	$[AGAT]_{14}(AGAC)_{10}$	Phillips et al. (2011)
24 (e)	170 bp	331 bp	270 bp	$[AGAT]_{18}(AGAC)_6$	M. Kline (unpublished)
24.2	172 bp	333 bp	272 bp	$[AGAT]_{12}AT[AGAT]_6(AGAC)_6$	Phillips et al. (2011)
24.3	173 bp	334 bp	273 bp	Not published	Hill et al. (2010)
25 (a)	174 bp	335 bp	274 bp	$[AGAT]_{16}(AGAC)_8AGAT$	Lareu et al. (1996)
25 (b)	174 bp	335 bp	274 bp	$[AGAT]_{16}(AGAC)_9$	Lareu et al. (1996)
25.2	176 bp	337 bp	276 bp	$[AGAT]_{12}AT[AGAT]_7(AGAC)_5AGAT$	Phillips et al. (2011)
26 (a)	178 bp	339 bp	278 bp	$[AGAT]_{17}(AGAC)_8AGAT$	Lareu et al. (1996)
26 (b)	178 bp	339 bp	278 bp	$[AGAT]_{17}(AGAC)_9$	Lareu et al. (1996)
26.2 (a)	180 bp	341 bp	280 bp	$[AGAT]_{12}AT[AGAT]_8(AGAC)_5AGAT$	Phillips et al. (2011)
26.2 (b)	180 bp	341 bp	280 bp	$[AGAT]_{14}AT[AGAT]_6(AGAC)_5AGAT$	Phillips et al. (2011)
27	182 bp	343 bp	282 bp	$[AGAT]_{18}(AGAC)_8AGAT$	Phillips et al. (2011)
27.2	184 bp	345 bp	284 bp	$[AGAT]_{15}AT[AGAT]_6(AGAC)_5AGAT$	Phillips et al. (2011)

52 observed alleles

D22S1045

GenBank Accession: AL022314 (allele 17). PCR product sizes of observed alleles

Allele (Repeat #)	Promega ESX 17	Promega ESI 17	ABI NGM	Repeat Structure $[ATT]_n$ ACT $[ATT]_2$	Reference
7	79 bp	306 bp	77 bp	Not published	Promega allelic ladders
8	82 bp	309 bp	80 bp	$[ATT]_5$ ACT $[ATT]_2$	Coble & Butler (2005)
9	85 bp	312 bp	83 bp	$[ATT]_6$ ACT $[ATT]_2$	Coble & Butler (2005)
10	88 bp	315 bp	86 bp	$[ATT]_7$ ACT $[ATT]_2$	Coble & Butler (2005)
11	91 bp	318 bp	89 bp	$[ATT]_8$ ACT $[ATT]_2$	Coble & Butler (2005)
12	94 bp	321 bp	92 bp	$[ATT]_9$ ACT $[ATT]_2$	Coble & Butler (2005)
13	97 bp	324 bp	95 bp	$[ATT]_{10}$ ACT $[ATT]_2$	Coble & Butler (2005)
14	100 bp	327 bp	98 bp	$[ATT]_{11}$ ACT $[ATT]_2$	Coble & Butler (2005)
15	103 bp	330 bp	101 bp	$[ATT]_{12}$ ACT $[ATT]_2$	Coble & Butler (2005)
16	106 bp	333 bp	104 bp	$[ATT]_{13}$ ACT $[ATT]_2$	Coble & Butler (2005)
17	109 bp	336 bp	107 bp	$[ATT]_{14}$ ACT $[ATT]_2$	Coble & Butler (2005)
18	112 bp	339 bp	110 bp	$[ATT]_{15}$ ACT $[ATT]_2$	Coble & Butler (2005)
19	115 bp	342 bp	113 bp	$[ATT]_{16}$ ACT $[ATT]_2$	Coble & Butler (2005)
20	118 bp	345 bp	116 bp	$[ATT]_{17}$ ACT $[ATT]_2$	Coble & Butler (2005)

14 observed alleles

SE33

GenBank Accession: V00481 (allele 26.2). PCR product sizes of observed alleles. As the SE33 repeat motif pattern is more complex than other STR loci and would take too much space to write out in full, the short-hand method of Rolf et al. (1997) has been adopted with the repeat motif pattern included across the top of the table and the number of repeats for each pattern indicated in the table. Thus, for allele 6.3, 2-1-3-1-7-3-1 means the flanking and central repeat sequence is [AAAG]2 AG [AAAG]3 AG [AAAG]7 [AAAG]3 AG. The light blue shaded AAAG repeat is the primary source of variation with the light yellow shaded AAAG repeat playing a role with larger alleles. PCR product sizes of observed alleles

				Repeat Motif Patterns																
Allele (Repeat #)	**ABI SEfiler**	**Promega ESX 17**	**Promega ESI 17**	AAAG	AG	AAAG	AG	AAAG	AAAAAG	AG	AGAAAG	**AAAG**	AAAAAG	AAAG	G	**AAGG**	AAAG/ANAG	AG	**Reference**
				5′ flanking				**central repeat**						3′ flanking					
3	197 bp	258 bp	300 bp	Not published															STRBase
4.2	203 bp	264 bp	306 bp	Not published															PowerPlex ESI allelic ladder
6.3	212 bp	273 bp	315 bp	2	1	3	1	7	0	0	0	0	0	0	0	0	3	1	Rolf et al. (1997)
7	213 bp	274 bp	316 bp	Not published															Lászik et al. (2001)
7.3	216 bp	277 bp	319 bp	2	1	3	1	8	0	0	0	0	0	0	0	0	3	1	Dauber et al. (2004)
8	217 bp	278 bp	320 bp	Not published															Poetsch et al. (2010)
8.1	218 bp	279 bp	321 bp	Not published															Lászik et al. (2001)
9 (a)	221 bp	282 bp	324 bp	2	1	3	1	9	0	0	0	0	0	0	1	0	3	1	Dauber et al. (2009)
9 (b)	221 bp	282 bp	324 bp	2	1	3	1	9	0	0	0	0	0	0	1	1	2	1	Kline et al. (2010)
9.2	223 bp	284 bp	326 bp	Not published															Lászik et al. (2001)
10	225 bp	286 bp	328 bp	Not published															Poetsch et al. (2011)
10.2	227 bp	288 bp	330 bp	2	1	0	0	18	0	0	0	0	0	0	1	0	3	1	Dauber et al. (2009)
10.3	228 bp	289 bp	331 bp	2	1	3	1	11	0	0	0	0	0	0	1	1	2	1	Urquhart et al. (1993)
11	229 bp	290 bp	332 bp	Not published															Poetsch et al. (2010)
11.1	230 bp	291 bp	333 bp	0	1	3	1	18	0	0	0	0	0	0	0	0	0	1	Urquhart et al. (1993)
11.2	231 bp	292 bp	334 bp	2	1	0	0	15	0	0	0	0	0	0	1	0	3	1	Dauber et al. (2004)

12	233 bp	294 bp	336 bp	2	1	3	1	12	0	0	0	0	0	0	1	0	3	1	Rolf et al. (1997)
12.2 (a)	235 bp	296 bp	338 bp	2	1	3	0	13	0	0	0	0	0	0	1	0	3	1	Rolf et al. (1997)
12.2 (b)	235 bp	296 bp	338 bp	0	1	3	1	16	0	0	0	0	0	0	1	0	3	1	Urquhart et al. (1993)
13	237 bp	298 bp	340 bp	2	1	3	1	13	0	0	0	0	0	0	1	0	3	1	Delghandi et al. (2001)
13.2	239 bp	300 bp	342 bp	2	1	3	0	14	0	0	0	0	0	0	1	0	3	1	Rolf et al. (1997); Kline et al. (2010)
13.3	240 bp	301 bp	343 bp	Not published															Poetsch et al. (2010)
14 (a)	241 bp	302 bp	344 bp	2	1	3	1	14	0	0	0	0	0	0	1	0	3	1	Rolf et al. (1997)
14 (b)	241 bp	302 bp	344 bp	2	1	3	1	14	0	0	0	0	0	0	1	1	2	1	Kline et al. (2010)
14.1	242 bp	303 bp	345 bp	Not published															Poetsch et al. (2010)
14.2	243 bp	304 bp	346 bp	2	1	3	0	15	0	0	0	0	0	0	1	0	3	1	Kline et al. (2010)
15	245 bp	306 bp	348 bp	2	1	3	1	15	0	0	0	0	0	0	1	0	3	1	Rolf et al. (1997)
15.2	247 bp	308 bp	350 bp	Not published															Lászik et al. (2001)
16 (a)	249 bp	310 bp	352 bp	2	1	3	1	16	0	0	0	0	0	0	1	0	3	1	Rolf et al. (1997)
16 (b)	249 bp	310 bp	352 bp	2	1	3	1	16	0	0	0	0	0	0	1	1	2	1	Kline et al. (2010)
16.1	250 bp	311 bp	353 bp	Not published															Berti et al. (2010)
16.2	251 bp	312 bp	354 bp	Not published															Lászik et al. (2001)
16.3	252 bp	313 bp	355 bp	Not published															Egyed et al. (2005)
17	253 bp	314 bp	356 bp	2	1	3	1	17	0	0	0	0	0	0	1	0	3	1	Rolf et al. (1997)
17.2	255 bp	316 bp	358 bp	Not published															Lászik et al. (2001)
17.3	256 bp	317 bp	359 bp	Not published															Lászik et al. (2001)
18	257 bp	318 bp	360 bp	2	1	3	1	18	0	0	0	0	0	0	1	0	3	1	Rolf et al. (1997)
18.2	259 bp	320 bp	362 bp	2	1	3	1	9	1	0	0	8	0	0	1	1	2	1	Dauber et al. (2009)
18.3	260 bp	321 bp	363 bp	2	1	0	0	19	0	0	0	0	+G	3	1	0	3	1	Urquhart et al. (1993)
19 (a)	261 bp	322 bp	364 bp	2	1	3	1	19	0	0	0	0	0	0	1	0	3	1	Rolf et al. (1997)

(Continued)

SE33 (*Continued*)

				Repeat Motif Patterns															
Allele (Repeat #)	**ABI SEfiler**	**Promega ESX 17**	**Promega ESI 17**	AAAG	AG	AAAG	AG	AAAG	AAAAAG	AG	AGAAAG	**AAAG**	AAAAAG	AAAG	G	**AAGG**	AAAG/ANAG	AG	**Reference**
19 (b)	261 bp	322 bp	364 bp	2	1	3	1	19	0	0	0	0	0	0	1	1	2	1	Kline et al. (2010)
19.2	263 bp	324 bp	366 bp	2	1	3	1	10	1	0	0	8	0	0	1	1	2	1	Rolf et al. (1997)
19.3	264 bp	325 bp	367 bp	Not published															Berti et al. (2010)
20 (a)	265 bp	326 bp	368 bp	2	1	3	1	20	0	0	0	0	0	0	1	0	3	1	Rolf et al. (1997)
20 (b)	265 bp	326 bp	368 bp	2	1	3	1	20	0	0	0	0	0	0	1	1	2	1	Kline et al. (2010)
20.2	267 bp	328 bp	370 bp	2	1	3	1	11	1	0	0	8	0	0	1	1	2	1	Rolf et al. (1997)
21	269 bp	330 bp	372 bp	2	1	3	1	21	0	0	0	0	0	0	1	0	3	1	Rolf et al. (1997)
21.2 (a)	271 bp	332 bp	374 bp	2	1	3	1	9	1	0	0	11	0	0	1	1	2	1	Rolf et al. (1997)
21.2 (b)	271 bp	332 bp	374 bp	2	1	3	1	11	1	0	0	9	0	0	1	1	2	1	Rolf et al. (1997)
21.2 (c)	271 bp	332 bp	374 bp	2	1	3	1	7	0	7	0	11	0	0	1	1	2	1	Kline et al. (2010)
22 (a)	273 bp	334 bp	376 bp	2	1	3	1	22	0	0	0	0	0	0	1	0	3	1	Rolf et al. (1997)
22 (b)	273 bp	334 bp	376 bp	2	1	3	1	21	0	0	0	0	0	0	1	1	3	1	Kline et al. (2010)
22.2 (a)	275 bp	336 bp	378 bp	2	1	3	1	7	1	0	0	14	0	0	1	1	2	1	Rolf et al. (1997)
22.2 (b)	275 bp	336 bp	378 bp	2	1	3	1	8	0	5	0	12	0	0	1	1	2	1	Rolf et al. (1997)
22.2 (c)	275 bp	336 bp	378 bp	2	1	3	1	9	1	0	0	12	0	0	1	1	2	1	Rolf et al. (1997)
22.2 (d)	275 bp	336 bp	378 bp	2	1	3	1	10	1	0	0	11	0	0	1	1	2	1	Rolf et al. (1997)
22.2 (e)	275 bp	336 bp	378 bp	2	1	3	1	11	1	0	0	10	0	0	1	1	2	1	Rolf et al. (1997)
22.2 (f)	275 bp	336 bp	378 bp	2	1	3	1	12	1	0	0	9	0	0	1	1	2	1	Rolf et al. (1997)
22.3	276 bp	337 bp	379 bp	Not published															Poetsch et al. (2010)
23	277 bp	338 bp	380 bp	Not published															Lászik et al. (2001)

23.2 (a)	279 bp	340 bp	382 bp	2	1	3	1	7	1	0	0	15	0	0	1	1	2	1	Rolf et al. (1997)
23.2 (b)	279 bp	340 bp	382 bp	2	1	3	1	8	1	0	0	14	0	0	1	1	2	1	Rolf et al. (1997)
23.2 (c)	279 bp	340 bp	382 bp	2	1	3	1	9	1	0	0	13	0	0	1	1	2	1	Rolf et al. (1997)
23.2 (d)	279 bp	340 bp	382 bp	2	1	3	1	10	0	3	0	12	0	0	1	1	2	1	Rolf et al. (1997)
23.2 (e)	279 bp	340 bp	382 bp	2	1	3	1	10	1	0	0	12	0	0	1	1	2	1	Rolf et al. (1997)
23.2 (f)	279 bp	340 bp	382 bp	2	1	3	1	11	1	0	0	11	0	0	1	1	2	1	Rolf et al. (1997)
23.2 (g)	279 bp	340 bp	382 bp	2	1	3	1	12	1	0	0	10	0	0	1	1	2	1	Rolf et al. (1997)
24	281 bp	342 bp	384 bp	Not published															Lászik et al. (2001)
24.2 (a)	283 bp	344 bp	386 bp	2	1	3	1	5	1	0	0	18	0	0	1	1	2	1	Rolf et al. (1997)
24.2 (b)	283 bp	344 bp	386 bp	2	1	3	1	7	1	0	0	16	0	0	1	1	2	1	Rolf et al. (1997)
24.2 (c)	283 bp	344 bp	386 bp	2	1	3	1	8	1	0	0	15	0	0	1	1	2	1	Rolf et al. (1997)
24.2 (d)	283 bp	344 bp	386 bp	2	1	3	1	9	1	0	0	14	0	0	1	1	2	1	Dauber et al. (2008)
24.2 (e)	283 bp	344 bp	386 bp	2	1	3	1	10	1	0	0	13	0	0	1	1	2	1	Rolf et al. (1997)
24.2 (f)	283 bp	344 bp	386 bp	2	1	3	1	11	1	0	0	12	0	0	1	1	2	1	Rolf et al. (1997)
24.2 (g)	283 bp	344 bp	386 bp	2	1	3	1	12	1	0	0	11	0	0	1	1	2	1	Rolf et al. (1997)
24.2 (h)	283 bp	344 bp	386 bp	2	1	3	1	6	1	0	0	16	0	0	1	1	3	1	U134C→T – Kline et al. (2010)
25	285 bp	346 bp	388 bp	Not published															Lászik et al. (2001)
25.2 (a)	287 bp	348 bp	390 bp	2	1	3	1	8	1	0	0	16	0	0	1	1	2	1	Delghandi et al. (2001)
25.2 (b)	287 bp	348 bp	390 bp	2	1	3	1	9	1	0	0	15	0	0	1	1	2	1	Rolf et al. (1997)
25.2 (c)	287 bp	348 bp	390 bp	2	1	3	1	10	0	0	1	14	0	0	1	1	2	1	Rolf et al. (1997)
25.2 (d)	287 bp	348 bp	390 bp	2	1	3	1	10	1	0	0	14	0	0	1	1	2	1	Rolf et al. (1997)
25.2 (e)	287 bp	348 bp	390 bp	2	1	3	1	11	1	0	0	13	0	0	1	1	2	1	Rolf et al. (1997)
25.2 (f)	287 bp	348 bp	390 bp	2	1	3	1	12	1	0	0	12	0	0	1	1	2	1	Rolf et al. (1997)
25.2 (g)	287 bp	348 bp	390 bp	2	1	3	1	14	1	0	0	10	0	0	1	1	2	1	Rolf et al. (1997)
25.2 (h)	287 bp	348 bp	390 bp	2	1	3	1	9	1	0	0	15	0	0	1	1	2	1	D75C→T – Kline et al. (2010)

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SE33 (*Continued*)

				Repeat Motif Patterns															
Allele (Repeat #)	**ABI SEfiler**	**Promega ESX 17**	**Promega ESI 17**	AAAG	AG	AAAG	AG	AAAG	AAAAAG	AG	AGAAAG	**AAAG**	AAAAAG	AAAG	G	**AAGG**	AAAG/ANAG	AG	**Reference**
25.2 (i)	287 bp	348 bp	390 bp	2	1	3	1	6	1	0	0	17	0	0	1	2	2	1	U134C→T – Kline et al. (2010)
25.3	288 bp	349 bp	391 bp	Not published															Berti et al. (2010)
26	289 bp	350 bp	392 bp	Not published															Lászik et al. (2001)
26.2 (a)	291 bp	352 bp	394 bp	2	1	3	1	8	1	0	0	17	0	0	1	1	2	1	Rolf et al. (1997)
26.2 (b)	291 bp	352 bp	394 bp	2	1	3	1	9	1	0	0	16	0	0	1	1	2	1	Rolf et al. (1997)
26.2 (c)	291 bp	352 bp	394 bp	2	1	3	1	10	1	0	0	15	0	0	1	1	2	1	Rolf et al. (1997)
26.2 (d)	291 bp	352 bp	394 bp	2	1	3	1	11	0	0	1	14	0	0	1	1	2	1	Rolf et al. (1997)
26.2 (e)	291 bp	352 bp	394 bp	2	1	3	1	11	1	0	0	14	0	0	1	1	2	1	Rolf et al. (1997)
26.2 (f)	291 bp	352 bp	394 bp	2	1	3	1	14	1	0	0	11	0	0	1	1	2	1	Rolf et al. (1997)
26.2 (g)	291 bp	352 bp	394 bp	2	1	3	1	6	1	0	0	18	0	0	1	1	3	1	U134C→T – Kline et al. (2010)
27	293 bp	354 bp	396 bp	Not published															Lászik et al. (2001)
27.2 (a)	295 bp	356 bp	398 bp	2	1	3	1	8	0	0	1	18	0	0	1	1	2	1	Rolf et al. (1997)
27.2 (b)	295 bp	356 bp	398 bp	2	1	3	1	8	1	0	0	18	0	0	1	1	2	1	Rolf et al. (1997)
27.2 (c)	295 bp	356 bp	398 bp	2	1	3	1	9	0	0	1	17	0	0	1	3	0	1	Rolf et al. (1997)
27.2 (d)	295 bp	356 bp	398 bp	2	1	3	1	10	1	0	0	16	0	0	1	1	2	1	Rolf et al. (1997)
27.2 (e)	295 bp	356 bp	398 bp	2	1	3	1	11	1	0	0	15	0	0	1	1	2	1	Rolf et al. (1997)
27.2 (f)	295 bp	356 bp	398 bp	2	1	3	1	12	1	0	0	14	0	0	1	1	2	1	Rolf et al. (1997)
27.2 (g)	295 bp	356 bp	398 bp	2	1	3	1	12	1	0	0	15	0	0	1	1	2	1	Rolf et al. (1997)
27.2 (h)	295 bp	356 bp	398 bp	2	1	3	1	13	0	0	1	13	0	0	1	3	0	1	Rolf et al. (1997)
27.2 (i)	295 bp	356 bp	398 bp	2	1	3	1	13	1	0	0	13	0	0	1	1	2	1	Rolf et al. (1997)
27.2 (j)	295 bp	356 bp	398 bp	2	1	3	1	15	1	0	0	11	0	0	1	1	2	1	Rolf et al. (1997)

27.2 (h)	295 bp	356 bp	398 bp	2	1	3	1	7	0	9	0	15	0	0	1	1	3	1	U134C→T – Kline et al. (2010)
27.3	296 bp	357 bp	399 bp	Not published															Hill et al. (2010)
28	297 bp	358 bp	400 bp	2	1	0	0	14	1	0	0	16	0	0	1	1	2	1	Dauber et al. (2009)
28.2 (a)	299 bp	360 bp	402 bp	2	1	3	1	8	1	0	0	19	0	0	1	1	2	1	Rolf et al. (1997)
28.2 (b)	299 bp	360 bp	402 bp	2	1	3	1	9	0	0	0	18	0	0	1	1	2	1	Rolf et al. (1997)
28.2 (c)	299 bp	360 bp	402 bp	2	1	3	1	9	0	0	0	15	0	0	1	1	2	1	Rolf et al. (1997)
28.2 (d)	299 bp	360 bp	402 bp	2	1	3	1	9	1	0	0	18	0	0	1	1	2	1	Rolf et al. (1997)
28.2 (e)	299 bp	360 bp	402 bp	2	1	3	1	10	1	0	0	17	0	0	1	1	2	1	Rolf et al. (1997)
28.2 (f)	299 bp	360 bp	402 bp	2	1	3	1	11	1	0	0	16	0	0	1	1	2	1	Rolf et al. (1997)
28.2 (g)	299 bp	360 bp	402 bp	2	1	3	1	12	1	0	0	15	0	0	1	1	2	1	Rolf et al. (1997)
28.2 (h)	299 bp	360 bp	402 bp	2	1	3	1	13	1	0	0	14	0	0	1	1	2	1	Rolf et al. (1997)
28.2 (i)	299 bp	360 bp	402 bp	2	1	3	1	14	1	0	0	13	0	0	1	1	2	1	Rolf et al. (1997)
28.2 (j)	299 bp	360 bp	402 bp	2	1	3	1	14	1	0	0	13	0	0	1	3	0	1	Rolf et al. (1997)
28.2 (k)	299 bp	360 bp	402 bp	2	1	3	1	16	1	0	0	11	0	0	1	1	2	1	Rolf et al. (1997)
28.3	300 bp	361 bp	403 bp	2	1	3	1	10	1	0	0	12	+A	4	1	1	2	1	Dauber et al. (2009)
29	301 bp	362 bp	404 bp	2	1	0	0	15	1	0	0	16	0	0	1	1	2	1	Dauber et al. (2009)
29.2 (a)	303 bp	364 bp	406 bp	2	1	3	1	8	1	0	0	20	0	0	1	1	2	1	Rolf et al. (1997)
29.2 (b)	303 bp	364 bp	406 bp	2	1	3	1	9	0	0	1	19	0	0	1	1	2	1	Rolf et al. (1997)
29.2 (c)	303 bp	364 bp	406 bp	2	1	3	1	9	1	0	0	19	0	0	1	1	2	1	Rolf et al. (1997)
29.2 (d)	303 bp	364 bp	406 bp	1	1	3	1	10	1	0	0	19	0	0	1	1	2	1	Rolf et al. (1997)
29.2 (e)	303 bp	364 bp	406 bp	2	1	3	1	11	0	5	0	16	0	0	1	1	2	1	Rolf et al. (1997)
29.2 (f)	303 bp	364 bp	406 bp	1	1	3	1	11	1	0	0	18	0	0	1	1	2	1	Rolf et al. (1997)
29.2 (g)	303 bp	364 bp	406 bp	1	1	3	1	11	0	0	1	18	0	0	1	1	2	1	Urquhart et al. (1993)
29.2 (h)	303 bp	364 bp	406 bp	2	1	3	1	11	1	0	0	17	0	0	1	1	2	1	Rolf et al. (1997)
29.2 (i)	303 bp	364 bp	406 bp	2	1	3	1	12	1	0	0	16	0	0	1	1	2	1	Rolf et al. (1997)

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SE33 (*Continued*)

				Repeat Motif Patterns															
Allele (Repeat #)	**ABI SEfiler**	**Promega ESX 17**	**Promega ESI 17**	AAAG	AG	AAAG	AG	AAAG	AAAAAG	AG	AGAAAG	**AAAG**	AAAAAG	AAAG	G	**AAGG**	AAAG/ANAG	AG	**Reference**
29.2 (j)	303 bp	364 bp	406 bp	2	1	3	1	13	0	0	1	15	0	0	1	3	0	1	Rolf et al. (1997)
29.2 (k)	303 bp	364 bp	406 bp	2	1	3	1	13	1	0	0	15	0	0	1	1	2	1	Rolf et al. (1997)
29.2 (l)	303 bp	364 bp	406 bp	2	1	3	1	14	1	0	0	14	0	0	1	1	2	1	Rolf et al. (1997)
29.2 (m)	303 bp	364 bp	406 bp	2	1	3	1	15	1	0	0	13	0	0	1	1	2	1	Delghandi et al. (2001)
29.2 (n)	303 bp	364 bp	406 bp	2	1	3	1	16	1	0	0	12	0	0	1	1	2	1	Rolf et al. (1997)
29.2 (o)	303 bp	364 bp	406 bp	2	1	3	1	11	1	0	0	17	0	0	1	1	2	1	D41-TTG-deletion – Kline et al. (2010)
29.3	304 bp	365 bp	407 bp	Not published															Hill et al. (2010)
30	305 bp	366 bp	408 bp	Not published															Lászik et al. (2001)
30.2 (a)	307 bp	368 bp	410 bp	2	1	3	1	11	1	0	0	18	0	0	1	1	2	1	Rolf et al. (1997)
30.2 (b)	307 bp	368 bp	410 bp	2	1	3	1	12	1	0	0	17	0	0	1	1	2	1	Rolf et al. (1997)
30.2 (c)	307 bp	368 bp	410 bp	1	1	3	1	12	1	0	0	18	0	0	1	1	2	1	Rolf et al. (1997)
30.2 (d)	307 bp	368 bp	410 bp	2	1	3	1	13	1	0	0	16	0	0	1	1	2	1	Rolf et al. (1997)
30.2 (e)	307 bp	368 bp	410 bp	2	1	3	1	14	1	0	0	15	0	0	1	1	2	1	Rolf et al. (1997)
30.2 (f)	307 bp	368 bp	410 bp	2	1	3	1	15	1	0	0	14	0	0	1	1	2	1	Rolf et al. (1997)
31	309 bp	370 bp	412 bp	Not published															Lászik et al. (2001)
31.2 (a)	311 bp	372 bp	414 bp	1	1	3	1	9	1	0	0	22	0	0	1	1	2	1	Rolf et al. (1997)
31.2 (b)	311 bp	372 bp	414 bp	1	1	3	1	10	1	0	0	21	0	0	1	1	2	1	Rolf et al. (1997)
31.2 (c)	311 bp	372 bp	414 bp	2	1	3	1	12	1	0	0	18	0	0	1	1	2	1	Rolf et al. (1997)

31.2 (d)	311 bp	372 bp	414 bp	2	1	3	1	13	1	0	0	17	0	0	1	1	2	1	Rolf et al. (1997)
31.2 (e)	311 bp	372 bp	414 bp	2	1	3	1	14	1	0	0	16	0	0	1	1	2	1	Rolf et al. (1997)
32	313 bp	374 bp	416 bp	2	1	3	1	11	1	0	0	9	1	9	1	1	2	1	Dauber et al. (2008)
32.2 (a)	315 bp	376 bp	418 bp	2	1	3	1	13	1	0	0	18	0	0	1	1	2	1	Rolf et al. (1997)
32.2 (b)	315 bp	376 bp	418 bp	2	1	3	1	14	1	0	0	17	0	0	1	1	2	1	Rolf et al. (1997)
33 (a)	317 bp	378 bp	420 bp	2	1	2	1	10	1	0	0	12	1	9	1	1	2	1	Rolf et al. (1997)
33 (b)	317 bp	378 bp	420 bp	2	1	3	1	10	1	0	0	11	1	9	1	1	2	1	Rolf et al. (1997)
33.2 (a)	319 bp	380 bp	422 bp	1	1	3	1	10	1	0	0	23	0	0	1	1	2	1	Rolf et al. (1997)
33.2 (b)	319 bp	380 bp	422 bp	2	1	3	1	12	1	0	0	20	0	0	1	1	2	1	Dauber et al. (2004)
33.2 (c)	319 bp	380 bp	422 bp	2	1	3	1	14	1	0	0	18	0	0	1	1	2	1	Delghandi et al. (2001)
34 (a)	321 bp	382 bp	424 bp	2	1	3	1	9	1	0	0	13	1	9	1	1	2	1	Rolf et al. (1997)
34 (b)	321 bp	382 bp	424 bp	2	1	3	1	10	1	0	0	12	1	9	1	1	2	1	Delghandi et al. (2001)
34 (c)	321 bp	382 bp	424 bp	2	1	3	1	11	1	0	0	11	1	9	1	1	2	1	Dauber et al. (2008)
34 (d)	321 bp	382 bp	424 bp	2	1	3	1	13	1	0	0	9	1	9	1	1	2	1	Delghandi et al. (2001)
34.2	323 bp	384 bp	426 bp	2	1	3	1	13	1	0	0	20	0	0	1	1	2	1	Dauber et al. (2004)
35	325 bp	386 bp	428 bp	Not published															Poetsch et al. (2010)
35.2 (a)	327 bp	388 bp	430 bp	1	1	3	1	13	1	0	0	22	0	0	1	1	2	1	Rolf et al. (1997)
35.2 (b)	327 bp	388 bp	430 bp	2	1	3	1	13	0	5	0	20	0	0	1	1	1	3	Urquhart et al. (1993)
36 (a)	329 bp	390 bp	432 bp	2	1	3	1	10	1	0	0	14	1	9	1	1	2	1	Rolf et al. (1997)
36 (b)	329 bp	390 bp	432 bp	2	1	3	1	11	1	0	0	5	1	17	1	1	2	1	Delghandi et al. (2001)
36 (c)	329 bp	390 bp	432 bp	2	1	3	1	12	1	0	0	10	1	11	1	1	2	1	Dauber et al. (2009)
36.2	331 bp	392 bp	434 bp	Not published															Lászik et al. (2001)
37 (a)	333 bp	394 bp	436 bp	2	1	3	1	9	1	0	0	16	1	9	1	1	2	1	Rolf et al. (1997)
37 (b)	333 bp	394 bp	436 bp	2	1	3	1	12	1	0	0	11	1	11	1	1	2	1	Dauber et al. (2008)
39	341 bp	402 bp	444 bp	Not published															PowerPlex ESI allelic ladder

(Continued)

SE33 (*Continued*)

				Repeat Motif Patterns																
Allele (Repeat #)	**ABI SEfiler**	**Promega ESX 17**	**Promega ESI 17**	AAAG	AG	AAAG	AG	AAAG	AAAAAG	AG	AGAAAG	**AAAG**	AAAAAG	AAAG	G	**AAGG**	AAAG/ANAG	AG	**Reference**
39.2	343 bp	404 bp	446 bp	Not published															Lászik et al. (2001)
41	349 bp	410 bp	452 bp	2	1	3	1	12	1	0	0	12	1	14	1	1	2	1	Lederer et al. (2008)
42	353 bp	414 bp	456 bp	Not published															PowerPlex ESI allelic ladder
49	381 bp	442 bp	484 bp	2	1	3	1	14	1	0	0	16	1	16	1	1	2	1	Klein et al. (2003)

178 observed alleles

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